

Mathematics of Machine Learning

Vector Spaces

$$Mx=b$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \\ 1 & 3 & 9 & 27 & 81 \\ 1 & 4 & 16 & 64 & 256 \\ 1 & 5 & 25 & 125 & 625 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \\ -3 \\ 1 \end{pmatrix}$$

$$b=M^{-1}x$$

$$Mv_i = \lambda_i v_i$$

$$\text{Tr}(X^{-1}A) = -X^{-1}AX^{-1}$$

Vectors, metric spaces, norms

Outline

1. **Vector Spaces**
2. Metrics (distances)
3. Norms

$$Mx=b$$

$$Mv_i = \lambda_i v_i$$

$$b=M \cdot$$

Matrices, Vectors and All That

- The language of machine learning is mathematics
- Sometimes we draw pretty pictures to explain the mathematics
- Much of the mathematics we will use involves vectors, matrices and functions
- You need to master the language of mathematics, otherwise you won't understand the algorithms
- I'm going to spend this lecture and the next revising the mathematics you need to know

Matrices, Vectors and All That

- The language of machine learning is mathematics
- Sometimes we draw pretty pictures to explain the mathematics
- Much of the mathematics we will use involves vectors, matrices and functions
- You need to master the language of mathematics, otherwise you won't understand the algorithms
- I'm going to spend this lecture and the next revising the mathematics you need to know

Matrices, Vectors and All That

- The language of machine learning is mathematics
- Sometimes we draw pretty pictures to explain the mathematics
- Much of the mathematics we will use involves vectors, matrices and functions
- You need to master the language of mathematics, otherwise you won't understand the algorithms
- I'm going to spend this lecture and the next revising the mathematics you need to know

Matrices, Vectors and All That

- The language of machine learning is mathematics
- Sometimes we draw pretty pictures to explain the mathematics
- Much of the mathematics we will use involves vectors, matrices and functions
- You need to master the language of mathematics, otherwise you won't understand the algorithms
- I'm going to spend this lecture and the next revising the mathematics you need to know

Matrices, Vectors and All That

- The language of machine learning is mathematics
- Sometimes we draw pretty pictures to explain the mathematics
- Much of the mathematics we will use involves vectors, matrices and functions
- You need to master the language of mathematics, otherwise you won't understand the algorithms
- I'm going to spend this lecture and the next revising the mathematics you need to know

Matrices, Vectors and All That

- The language of machine learning is mathematics
- Sometimes we draw pretty pictures to explain the mathematics
- Much of the mathematics we will use involves vectors, matrices and functions
- You need to master the language of mathematics, otherwise you won't understand the algorithms
- I'm going to spend this lecture and the next revising the mathematics you need to know (but I'm going use a slightly posher language than you are probably used to)

Scalars (Fields)

- Vector spaces involve **fields** (numbers)
- These are quantities we can add together ($a + b$) and multiply together ($a \times b$)

Scalars (Fields)

- Vector spaces involve **fields** (numbers)—aka **scalars**
- These are quantities we can add together ($a + b$) and multiply together ($a \times b$)

Scalars (Fields)

- Vector spaces involve **fields** (numbers)—aka **scalars**
- These are quantities we can add together ($a + b$) and multiply together ($a \times b$)

Scalars (Fields)

- Vector spaces involve **fields** (numbers)—aka **scalars**
- These are quantities we can add together ($a + b$) and multiply together ($a \times b$)
- Formally they form an Abelian group under addition with an identity 0

Scalars (Fields)

- Vector spaces involve **fields** (numbers)—aka **scalars**
- These are quantities we can add together ($a + b$) and multiply together ($a \times b$)
- Formally they form an Abelian group under addition with an identity 0 and excluding 0 an Abelian group under multiplication

Scalars (Fields)

- Vector spaces involve **fields** (numbers)—aka **scalars**
- These are quantities we can add together ($a + b$) and multiply together ($a \times b$)
- Formally they form an Abelian group under addition with an identity 0 and excluding 0 an Abelian group under multiplication and they are distributive

$$a \times (b + c) = a \times b + a \times c$$

Scalars (Fields)

- Vector spaces involve **fields** (numbers)—aka **scalars**
- These are quantities we can add together ($a + b$) and multiply together ($a \times b$)
- Formally they form an Abelian group under addition with an identity 0 and excluding 0 an Abelian group under multiplication and they are distributive

$$a \times (b + c) = a \times b + a \times c$$

- Although this sounds rather daunting don't panic

Scalars (Fields)

- Vector spaces involve **fields** (numbers)—aka **scalars**
- These are quantities we can add together ($a + b$) and multiply together ($a \times b$)
- Formally they form an Abelian group under addition with an identity 0 and excluding 0 an Abelian group under multiplication and they are distributive

$$a \times (b + c) = a \times b + a \times c$$

- Although this sounds rather daunting don't panic. They behave like numbers.

Scalars (Fields)

- Vector spaces involve **fields** (numbers)—aka **scalars**
- These are quantities we can add together ($a + b$) and multiply together ($a \times b$)
- Formally they form an Abelian group under addition with an identity 0 and excluding 0 an Abelian group under multiplication and they are distributive

$$a \times (b + c) = a \times b + a \times c$$

- Although this sounds rather daunting don't panic. They behave like numbers. The field might be integers, rational numbers, reals, complex numbers or something a bit more exotic

Scalars (Fields)

- Vector spaces involve **fields** (numbers)—aka **scalars**
- These are quantities we can add together ($a + b$) and multiply together ($a \times b$)
- Formally they form an Abelian group under addition with an identity 0 and excluding 0 an Abelian group under multiplication and they are distributive

$$a \times (b + c) = a \times b + a \times c$$

- Although this sounds rather daunting don't panic. They behave like numbers. The field might be integers, rational numbers, reals, complex numbers or something a bit more exotic—but we will almost always consider reals

Vectors

- We often work with objects with many components (features)
- To help handle this we will use vector notation
 - We represent vectors by bold symbols
 - All our vectors are column vectors by default
 - We treat them as $n \times 1$ matrix
- We write row vectors as transposes of column vectors

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$\mathbf{y}^T = (y_1, y_2, \dots, y_n)$$

Vectors

- We often work with objects with many components (features)
- To help handle this we will use vector notation

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

- We represent vectors by bold symbols
 - All our vectors are column vectors by default
 - We treat them as $n \times 1$ matrix
- We write row vectors as transposes of column vectors

$$\mathbf{y}^T = (y_1, y_2, \dots, y_n)$$

Vectors

- We often work with objects with many components (features)
- To help handle this we will use vector notation
 - We represent vectors by bold symbols
 - All our vectors are column vectors by default
 - We treat them as $n \times 1$ matrix
- We write row vectors as transposes of column vectors

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$\mathbf{y}^T = (y_1, y_2, \dots, y_n)$$

Vectors

- We often work with objects with many components (features)
- To help handle this we will use vector notation
- We represent vectors by bold symbols
- All our vectors are column vectors by default
- We treat them as $n \times 1$ matrix
- We write row vectors as transposes of column vectors

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$\mathbf{y}^T = (y_1, y_2, \dots, y_n)$$

Vectors

- We often work with objects with many components (features)
- To help handle this we will use vector notation
- We represent vectors by bold symbols
- All our vectors are column vectors by default
- We treat them as $n \times 1$ matrix
- We write row vectors as transposes of column vectors

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$\mathbf{y}^T = (y_1, y_2, \dots, y_n)$$

Vectors

- We often work with objects with many components (features)
- To help handle this we will use vector notation
- We represent vectors by bold symbols
- All our vectors are column vectors by default
- We treat them as $n \times 1$ matrix
- We write row vectors as transposes of column vectors

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$\mathbf{y}^T = (y_1, y_2, \dots, y_n)$$

Vectors

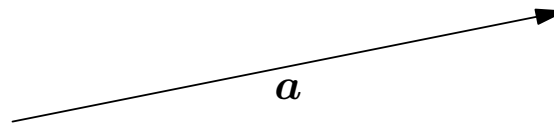
- We often work with objects with many components (features)
- To help handle this we will use vector notation
- We represent vectors by bold symbols
- All our vectors are column vectors by default
- We treat them as $n \times 1$ matrix
- We write row vectors as transposes of column vectors

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$\mathbf{y}^T = (y_1, y_2, \dots, y_n)$$

Basic Vector Operations

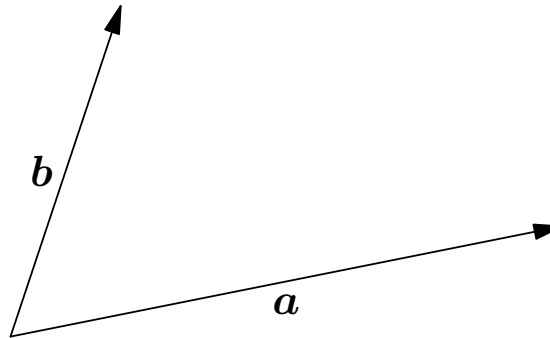
- The basic vector operations are adding



- multiplying by a scalar (a number)

Basic Vector Operations

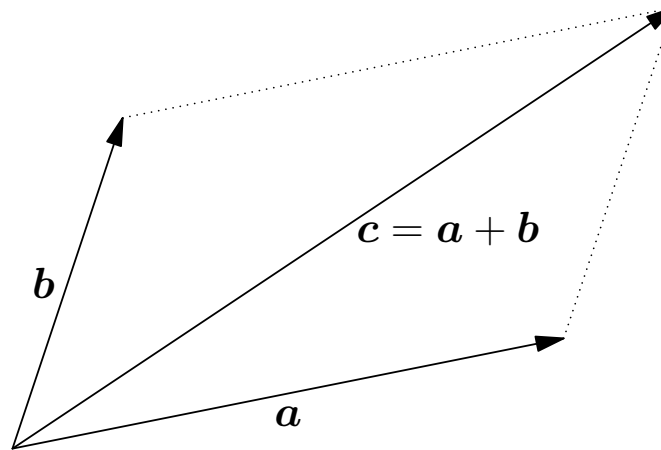
- The basic vector operations are adding



- multiplying by a scalar (a number)

Basic Vector Operations

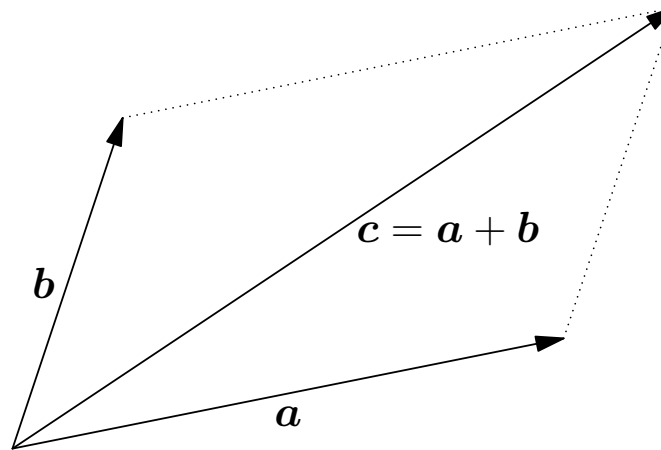
- The basic vector operations are adding



- multiplying by a scalar (a number)

Basic Vector Operations

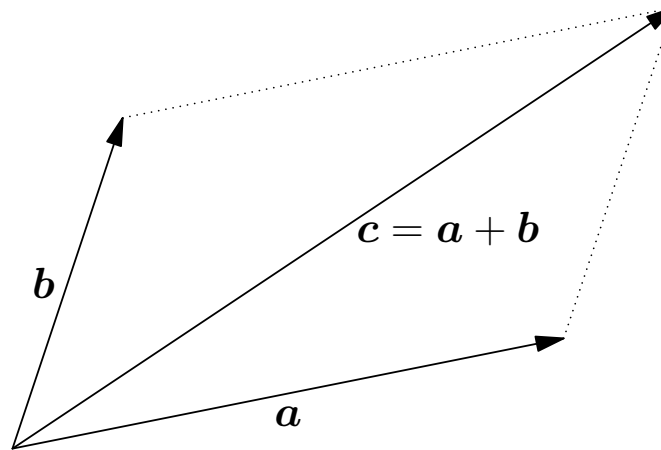
- The basic vector operations are adding



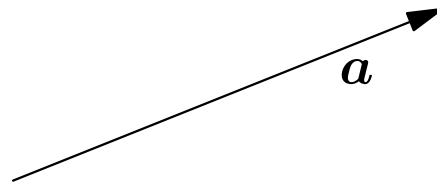
- multiplying by a scalar (a number)

Basic Vector Operations

- The basic vector operations are adding

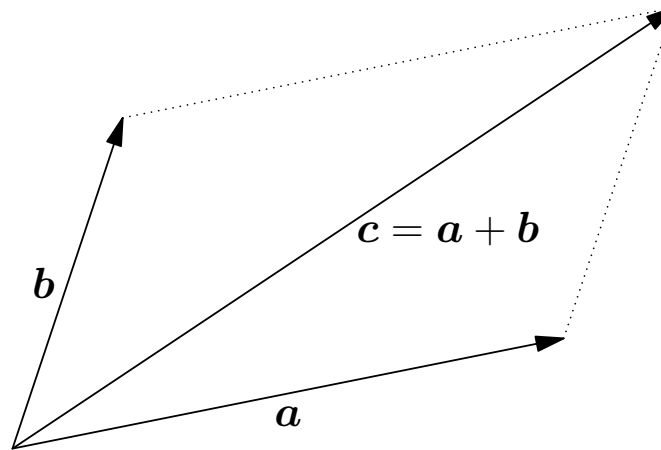


- multiplying by a scalar (a number)

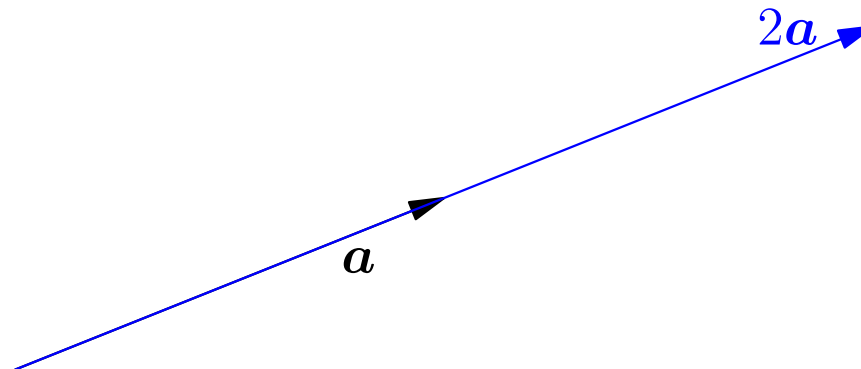


Basic Vector Operations

- The basic vector operations are adding



- multiplying by a scalar (a number)



Vector Space

- A vector space, $\mathcal{V}$, is a set of vectors which satisfies

1. if $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ then $a\mathbf{v} \in \mathcal{V}$ and $\mathbf{v} + \mathbf{w} \in \mathcal{V}$ (closure)
2. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity of addition)
3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity of addition)
4. $\mathbf{v} + \mathbf{0} = \mathbf{v}$ (existence of additive identity $\mathbf{0}$)
5. $1\mathbf{v} = \mathbf{v}$ (existence of multiplicative identity 1)
6. $a(b\mathbf{v}) = (ab)\mathbf{v}$ (distributive properties)
7. $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$
8. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

(You don't need to remember these)

- Just from these properties we can deduce other properties

Vector Space

- A vector space, $\mathcal{V}$, is a set of vectors which satisfies
 1. if $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ then $a\mathbf{v} \in \mathcal{V}$ and $\mathbf{v} + \mathbf{w} \in \mathcal{V}$ (closure)
 2. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity of addition)
 3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity of addition)
 4. $\mathbf{v} + \mathbf{0} = \mathbf{v}$ (existence of additive identity $\mathbf{0}$)
 5. $1\mathbf{v} = \mathbf{v}$ (existence of multiplicative identity 1)
 6. $a(b\mathbf{v}) = (ab)\mathbf{v}$ (distributive properties)
 7. $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$
 8. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

(You don't need to remember these)

- Just from these properties we can deduce other properties

Vector Space

- A vector space, $\mathcal{V}$, is a set of vectors which satisfies
 1. if $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ then $a\mathbf{v} \in \mathcal{V}$ and $\mathbf{v} + \mathbf{w} \in \mathcal{V}$ (closure)
 2. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity of addition)
 3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity of addition)
 4. $\mathbf{v} + \mathbf{0} = \mathbf{v}$ (existence of additive identity $\mathbf{0}$)
 5. $1\mathbf{v} = \mathbf{v}$ (existence of multiplicative identity 1)
 6. $a(b\mathbf{v}) = (ab)\mathbf{v}$ (distributive properties)
 7. $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$
 8. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

(You don't need to remember these)

- Just from these properties we can deduce other properties

Vector Space

- A vector space, $\mathcal{V}$, is a set of vectors which satisfies
 1. if $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ then $a\mathbf{v} \in \mathcal{V}$ and $\mathbf{v} + \mathbf{w} \in \mathcal{V}$ (closure)
 2. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity of addition)
 3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity of addition)
 4. $\mathbf{v} + \mathbf{0} = \mathbf{v}$ (existence of additive identity $\mathbf{0}$)
 5. $1\mathbf{v} = \mathbf{v}$ (existence of multiplicative identity 1)
 6. $a(b\mathbf{v}) = (ab)\mathbf{v}$ (distributive properties)
 7. $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$
 8. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

(You don't need to remember these)

- Just from these properties we can deduce other properties

Vector Space

- A vector space, $\mathcal{V}$, is a set of vectors which satisfies
 1. if $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ then $a\mathbf{v} \in \mathcal{V}$ and $\mathbf{v} + \mathbf{w} \in \mathcal{V}$ (closure)
 2. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity of addition)
 3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity of addition)
 4. $\mathbf{v} + \mathbf{0} = \mathbf{v}$ (existence of additive identity $\mathbf{0}$)
 5. $1\mathbf{v} = \mathbf{v}$ (existence of multiplicative identity 1)
 6. $a(b\mathbf{v}) = (ab)\mathbf{v}$ (distributive properties)
 7. $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$
 8. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

(You don't need to remember these)

- Just from these properties we can deduce other properties

Vector Space

- A vector space, $\mathcal{V}$, is a set of vectors which satisfies
 1. if $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ then $a\mathbf{v} \in \mathcal{V}$ and $\mathbf{v} + \mathbf{w} \in \mathcal{V}$ (closure)
 2. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity of addition)
 3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity of addition)
 4. $\mathbf{v} + \mathbf{0} = \mathbf{v}$ (existence of additive identity $\mathbf{0}$)
 5. $1\mathbf{v} = \mathbf{v}$ (existence of multiplicative identity 1)
 6. $a(b\mathbf{v}) = (ab)\mathbf{v}$ (distributive properties)
 7. $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$
 8. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

(You don't need to remember these)

- Just from these properties we can deduce other properties

Vector Space

- A vector space, $\mathcal{V}$, is a set of vectors which satisfies
 1. if $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ then $a\mathbf{v} \in \mathcal{V}$ and $\mathbf{v} + \mathbf{w} \in \mathcal{V}$ (closure)
 2. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity of addition)
 3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity of addition)
 4. $\mathbf{v} + \mathbf{0} = \mathbf{v}$ (existence of additive identity $\mathbf{0}$)
 5. $1\mathbf{v} = \mathbf{v}$ (existence of multiplicative identity 1)
 6. $a(b\mathbf{v}) = (ab)\mathbf{v}$ (distributive properties)
 7. $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$
 8. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

(You don't need to remember these)

- Just from these properties we can deduce other properties

Vector Space

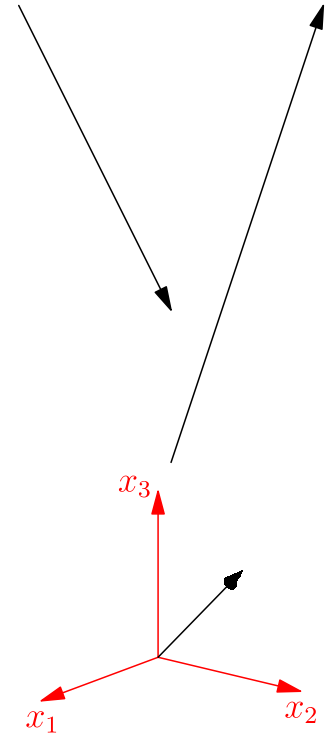
- A vector space, $\mathcal{V}$, is a set of vectors which satisfies
 1. if $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ then $a\mathbf{v} \in \mathcal{V}$ and $\mathbf{v} + \mathbf{w} \in \mathcal{V}$ (closure)
 2. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity of addition)
 3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity of addition)
 4. $\mathbf{v} + \mathbf{0} = \mathbf{v}$ (existence of additive identity $\mathbf{0}$)
 5. $1\mathbf{v} = \mathbf{v}$ (existence of multiplicative identity 1)
 6. $a(b\mathbf{v}) = (ab)\mathbf{v}$ (distributive properties)
 7. $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$
 8. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

(You don't need to remember these)

- Just from these properties we can deduce other properties

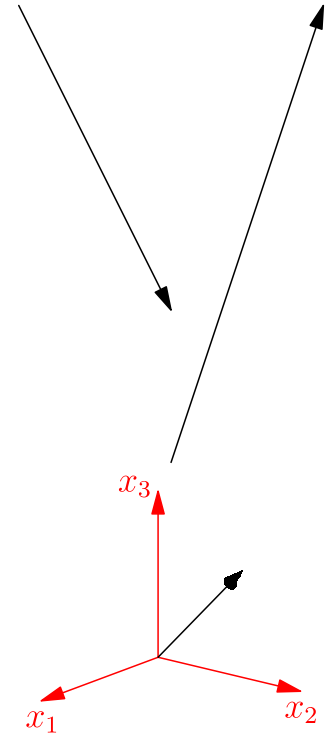
$$\mathbb{R}^n$$

- When we first learn about vectors we think of them as arrows in 3-D space
- If we centre them all at the origin then there is a one-to-one correspondence between vectors and points in space
- We call this vector space $\mathbb{R}^3$
- Any set of quantities $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$ which satisfy the axioms above form a vector space $\mathbb{R}^n$
- Of course, we can't so easily draw pictures of high-dimensional vectors



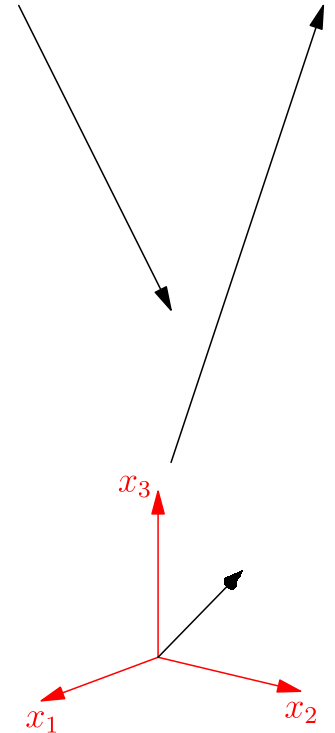
$$\mathbb{R}^n$$

- When we first learn about vectors we think of them as arrows in 3-D space
- If we centre them all at the origin then there is a one-to-one correspondence between vectors and points in space
- We call this vector space $\mathbb{R}^3$
- Any set of quantities $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$ which satisfy the axioms above form a vector space $\mathbb{R}^n$
- Of course, we can't so easily draw pictures of high-dimensional vectors



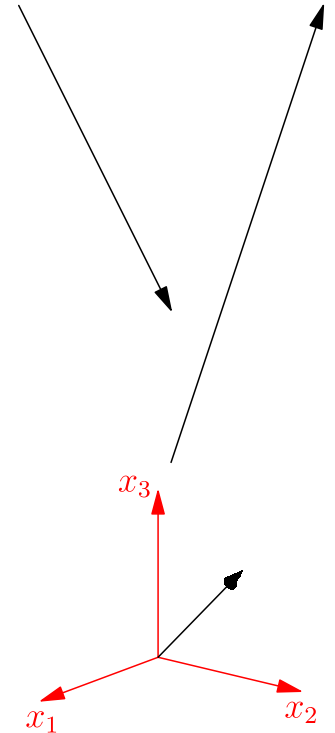
$$\mathbb{R}^n$$

- When we first learn about vectors we think of them as arrows in 3-D space
- If we centre them all at the origin then there is a one-to-one correspondence between vectors and points in space
- We call this vector space $\mathbb{R}^3$
- Any set of quantities $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$ which satisfy the axioms above form a vector space $\mathbb{R}^n$
- Of course, we can't so easily draw pictures of high-dimensional vectors



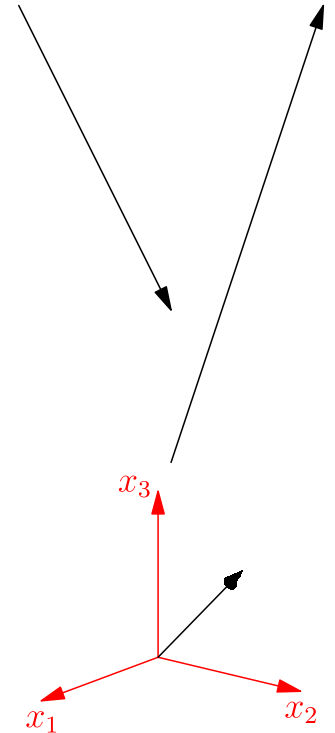
$$\mathbb{R}^n$$

- When we first learn about vectors we think of them as arrows in 3-D space
- If we centre them all at the origin then there is a one-to-one correspondence between vectors and points in space
- We call this vector space $\mathbb{R}^3$
- Any set of quantities $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$ which satisfy the axioms above form a vector space $\mathbb{R}^n$
- Of course, we can't so easily draw pictures of high-dimensional vectors



$$\mathbb{R}^n$$

- When we first learn about vectors we think of them as arrows in 3-D space
- If we centre them all at the origin then there is a one-to-one correspondence between vectors and points in space
- We call this vector space $\mathbb{R}^3$
- Any set of quantities $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$ which satisfy the axioms above form a vector space $\mathbb{R}^n$
- Of course, we can't so easily draw pictures of high-dimensional vectors



Other Vector Spaces

- Any set of object that satisfies the axioms of a vector spacer are *vectors*
- Matrices satisfy all the conditions of a vector space
- Infinite sequences form a vector space
- Functions form a vector space
 - ★ Let $C(a,b)$ be the set of functions defined on the interval $[a,b]$
 - ★ Note that if $f(x), g(x) \in C(a,b)$ then $af(x) \in C(a,b)$ and $f(x) + g(x) \in C(a,b)$
- Bounded vectors in $\mathbb{R}^n$ **don't** form a vector space

Other Vector Spaces

- Any set of object that satisfies the axioms of a vector spacer are *vectors*—not just $v \in \mathbb{R}^n$
- Matrices satisfy all the conditions of a vector space
- Infinite sequences form a vector space
- Functions form a vector space
 - ★ Let $C(a,b)$ be the set of functions defined on the interval $[a,b]$
 - ★ Note that if $f(x), g(x) \in C(a,b)$ then $af(x) \in C(a,b)$ and $f(x) + g(x) \in C(a,b)$
- Bounded vectors in $\mathbb{R}^n$ **don't** form a vector space

Other Vector Spaces

- Any set of object that satisfies the axioms of a vector spacer are *vectors*—not just $v \in \mathbb{R}^n$
- Matrices satisfy all the conditions of a vector space
- Infinite sequences form a vector space
- Functions form a vector space
 - ★ Let $C(a,b)$ be the set of functions defined on the interval $[a,b]$
 - ★ Note that if $f(x), g(x) \in C(a,b)$ then $af(x) \in C(a,b)$ and $f(x) + g(x) \in C(a,b)$
- Bounded vectors in $\mathbb{R}^n$ **don't** form a vector space

Other Vector Spaces

- Any set of object that satisfies the axioms of a vector spacer are *vectors*—not just $v \in \mathbb{R}^n$
- Matrices satisfy all the conditions of a vector space
- Infinite sequences form a vector space
- Functions form a vector space
 - ★ Let $C(a,b)$ be the set of functions defined on the interval $[a,b]$
 - ★ Note that if $f(x), g(x) \in C(a,b)$ then $af(x) \in C(a,b)$ and $f(x) + g(x) \in C(a,b)$
- Bounded vectors in $\mathbb{R}^n$ **don't** form a vector space

Other Vector Spaces

- Any set of object that satisfies the axioms of a vector spacer are *vectors*—not just $v \in \mathbb{R}^n$
- Matrices satisfy all the conditions of a vector space
- Infinite sequences form a vector space
- Functions form a vector space
 - ★ Let $C(a,b)$ be the set of functions defined on the interval $[a,b]$
 - ★ Note that if $f(x), g(x) \in C(a,b)$ then $af(x) \in C(a,b)$ and $f(x) + g(x) \in C(a,b)$
- Bounded vectors in $\mathbb{R}^n$ **don't** form a vector space

Other Vector Spaces

- Any set of object that satisfies the axioms of a vector spacer are *vectors*—not just $v \in \mathbb{R}^n$
- Matrices satisfy all the conditions of a vector space
- Infinite sequences form a vector space
- Functions form a vector space
 - ★ Let $C(a,b)$ be the set of functions defined on the interval $[a,b]$
 - ★ Note that if $f(x), g(x) \in C(a,b)$ then $af(x) \in C(a,b)$ and $f(x) + g(x) \in C(a,b)$
- Bounded vectors in $\mathbb{R}^n$ **don't** form a vector space

Other Vector Spaces

- Any set of object that satisfies the axioms of a vector spacer are *vectors*—not just $v \in \mathbb{R}^n$
- Matrices satisfy all the conditions of a vector space
- Infinite sequences form a vector space
- Functions form a vector space
 - ★ Let $C(a,b)$ be the set of functions defined on the interval $[a,b]$
 - ★ Note that if $f(x), g(x) \in C(a,b)$ then $af(x) \in C(a,b)$ and $f(x) + g(x) \in C(a,b)$
- Bounded vectors in $\mathbb{R}^n$ **don't** form a vector space

Outline

1. Vector Spaces
2. Metrics (distances)
3. Norms

$$Mx=b$$

$$Mv_i = \lambda_i v_i$$

$$b=M$$

Metrics

- Vector spaces become more interesting if we have a notion of distance
- We say $d(x, y)$ is a **proper distance** or **metric** if
 1. $d(x, y) \geq 0$ (non-negativity)
 2. $d(x, y) = 0$ iff $x = y$ (identity of indiscernibles)
 3. $d(x, y) = d(y, x)$ (symmetry)
 4. $d(x, y) \leq d(x, z) + d(z, y)$ (triangular inequality)
- There are typically many possible distances (e.g. Euclidean distance, Manhattan distance, etc.)
- Often one or more condition isn't satisfied then we have a **pseudo-metric**

Metrics

- Vector spaces become more interesting if we have a notion of distance
- We say $d(x, y)$ is a **proper distance** or **metric** if
 1. $d(x, y) \geq 0$ (non-negativity)
 2. $d(x, y) = 0$ iff $x = y$ (identity of indiscernibles)
 3. $d(x, y) = d(y, x)$ (symmetry)
 4. $d(x, y) \leq d(x, z) + d(z, y)$ (triangular inequality)
- There are typically many possible distances (e.g. Euclidean distance, Manhattan distance, etc.)
- Often one or more condition isn't satisfied then we have a **pseudo-metric**

Metrics

- Vector spaces become more interesting if we have a notion of distance
- We say $d(x, y)$ is a **proper distance** or **metric** if
 1. $d(x, y) \geq 0$ (non-negativity)
 2. $d(x, y) = 0$ iff $x = y$ (identity of indiscernibles)
 3. $d(x, y) = d(y, x)$ (symmetry)
 4. $d(x, y) \leq d(x, z) + d(z, y)$ (triangular inequality)
- There are typically many possible distances (e.g. Euclidean distance, Manhattan distance, etc.)
- Often one or more condition isn't satisfied then we have a **pseudo-metric**

Metrics

- Vector spaces become more interesting if we have a notion of distance
- We say $d(x, y)$ is a **proper distance** or **metric** if
 1. $d(x, y) \geq 0$ (non-negativity)
 2. $d(x, y) = 0$ iff $x = y$ (identity of indiscernibles)
 3. $d(x, y) = d(y, x)$ (symmetry)
 4. $d(x, y) \leq d(x, z) + d(z, y)$ (triangular inequality)
- There are typically many possible distances (e.g. Euclidean distance, Manhattan distance, etc.)
- Often one or more condition isn't satisfied then we have a **pseudo-metric**

Metrics

- Vector spaces become more interesting if we have a notion of distance
- We say $d(x, y)$ is a **proper distance** or **metric** if
 1. $d(x, y) \geq 0$ (non-negativity)
 2. $d(x, y) = 0$ iff $x = y$ (identity of indiscernibles)
 3. $d(x, y) = d(y, x)$ (symmetry)
 4. $d(x, y) \leq d(x, z) + d(z, y)$ (triangular inequality)
- There are typically many possible distances (e.g. Euclidean distance, Manhattan distance, etc.)
- Often one or more condition isn't satisfied then we have a **pseudo-metric**

Metrics

- Vector spaces become more interesting if we have a notion of distance
- We say $d(\mathbf{x}, \mathbf{y})$ is a **proper distance** or **metric** if
 1. $d(\mathbf{x}, \mathbf{y}) \geq 0$ (non-negativity)
 2. $d(\mathbf{x}, \mathbf{y}) = 0$ iff $\mathbf{x} = \mathbf{y}$ (identity of indiscernibles)
 3. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ (symmetry)
 4. $d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y})$ (triangular inequality)
- There are typically many possible distances (e.g. Euclidean distance, Manhattan distance, etc.)
- Often one or more condition isn't satisfied then we have a **pseudo-metric**

Metrics

- Vector spaces become more interesting if we have a notion of distance
- We say $d(\mathbf{x}, \mathbf{y})$ is a **proper distance** or **metric** if
 1. $d(\mathbf{x}, \mathbf{y}) \geq 0$ (non-negativity)
 2. $d(\mathbf{x}, \mathbf{y}) = 0$ iff $\mathbf{x} = \mathbf{y}$ (identity of indiscernibles)
 3. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ (symmetry)
 4. $d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y})$ (triangular inequality)
- There are typically many possible distances (e.g. Euclidean distance, Manhattan distance, etc.)
- Often one or more condition isn't satisfied then we have a **pseudo-metric**

Mappings and Functions

- A function defines a mapping from one vector space to another (although the spaces might be the same)

Mappings and Functions

- A function defines a mapping from one vector space to another (although the spaces might be the same), e.g.

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

Mappings and Functions

- A function defines a mapping from one vector space to another (although the spaces might be the same), e.g.

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

(f maps the reals onto reals, i.e. $f(x)$ takes a real x and gives you a new real number $y = f(x)$)

Mappings and Functions

- A function defines a mapping from one vector space to another (although the spaces might be the same), e.g.

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

(f maps the reals onto reals, i.e. $f(x)$ takes a real x and gives you a new real number $y = f(x)$)

- We are often interested in functions that behave nicely

Mappings and Functions

- A function defines a mapping from one vector space to another (although the spaces might be the same), e.g.

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

(f maps the reals onto reals, i.e. $f(x)$ takes a real x and gives you a new real number $y = f(x)$)

- We are often interested in functions that behave nicely
- E.g. They are continuous

Lipschitz Function

- One way to characterise well behaved function, $f(x)$ is if there exists a number $K < \infty$ such that for all x and y

$$d(f(x), f(y)) \leq K d(x, y)$$

- This is known as a **Lipschitz condition** and the function is said to be K -Lipschitz
- Note that such functions cannot have any jumps (i.e. they are continuous)
- The size of K measures the limit on the amplifying effect of the function

Lipschitz Function

- One way to characterise well behaved function, $f(x)$ is if there exists a number $K < \infty$ such that for all x and y

$$d(f(x), f(y)) \leq K d(x, y)$$

- This is known as a **Lipschitz condition** and the function is said to be K -Lipschitz
- Note that such functions cannot have any jumps (i.e. they are continuous)
- The size of K measures the limit on the amplifying effect of the function

Lipschitz Function

- One way to characterise well behaved function, $f(x)$ is if there exists a number $K < \infty$ such that for all x and y

$$d(f(x), f(y)) \leq K d(x, y)$$

- This is known as a **Lipschitz condition** and the function is said to be K -Lipschitz or Lipschitz continuous
- Note that such functions cannot have any jumps (i.e. they are continuous)
- The size of K measures the limit on the amplifying effect of the function

Lipschitz Function

- One way to characterise well behaved function, $f(x)$ is if there exists a number $K < \infty$ such that for all x and y

$$d(f(x), f(y)) \leq K d(x, y)$$

- This is known as a **Lipschitz condition** and the function is said to be K -Lipschitz or Lipschitz continuous
- Note that such functions cannot have any jumps (i.e. they are continuous)
- The size of K measures the limit on the amplifying effect of the function

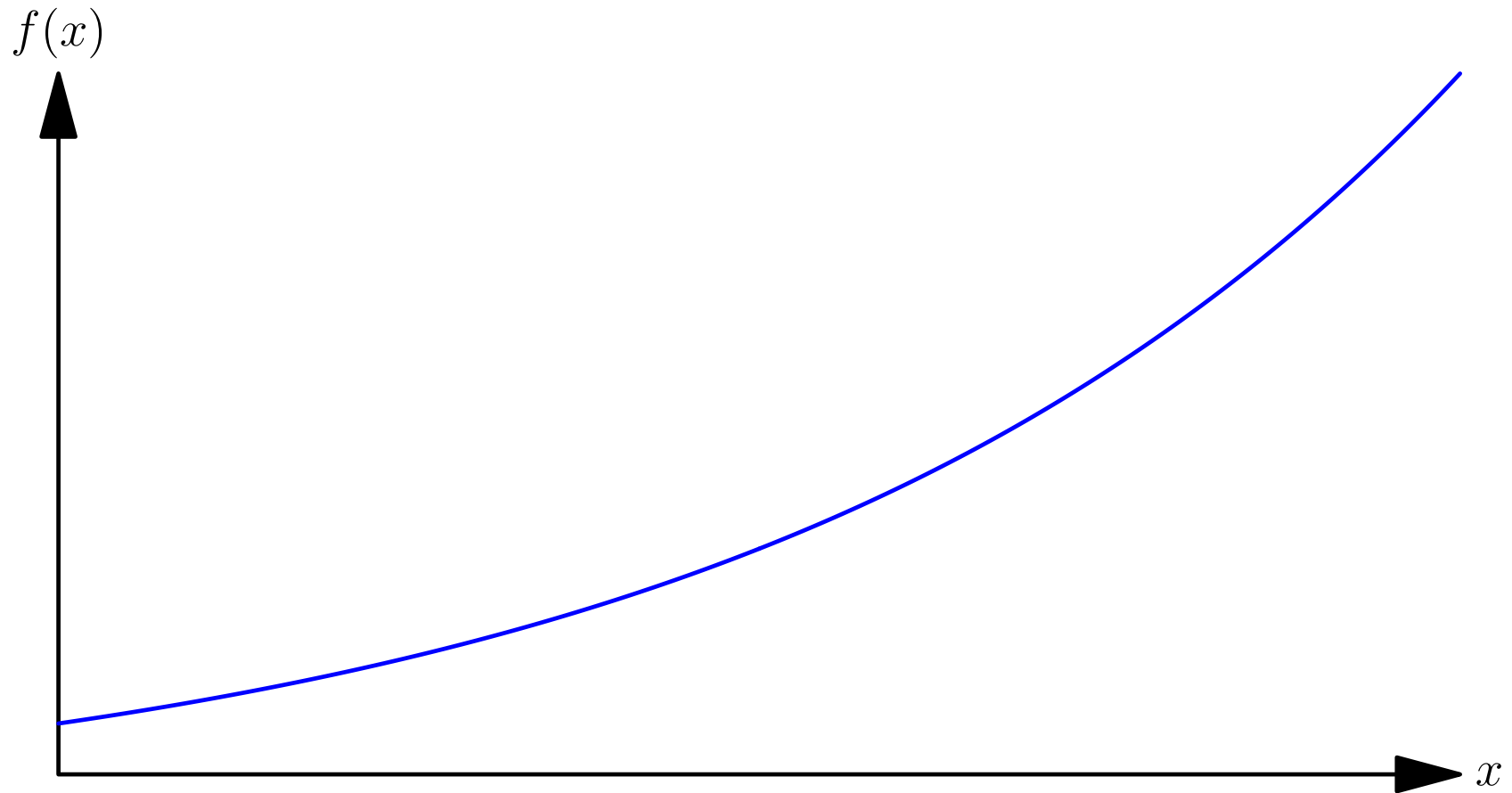
Lipschitz Function

- One way to characterise well behaved function, $f(x)$ is if there exists a number $K < \infty$ such that for all x and y

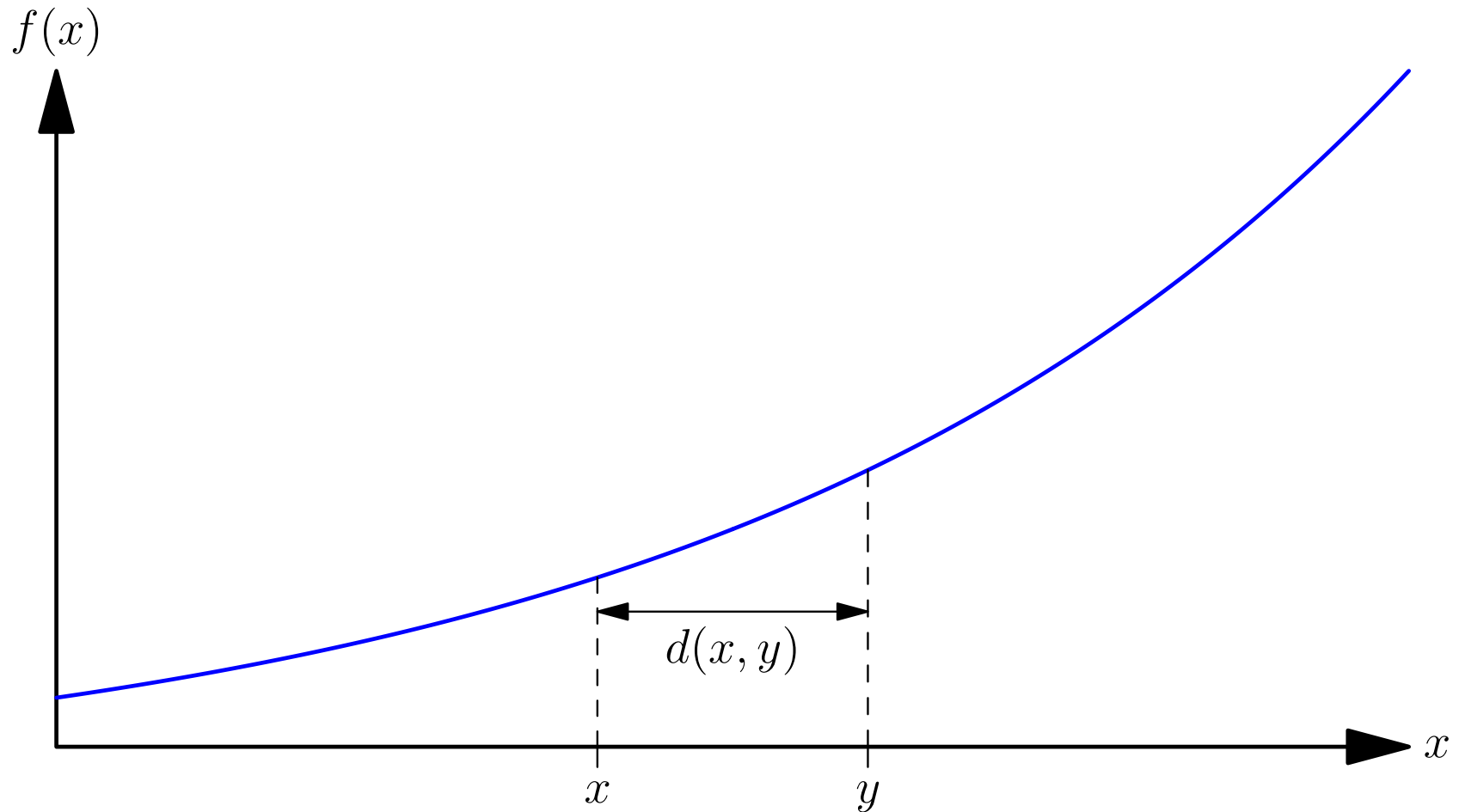
$$d(f(x), f(y)) \leq K d(x, y)$$

- This is known as a **Lipschitz condition** and the function is said to be K -Lipschitz or Lipschitz continuous
- Note that such functions cannot have any jumps (i.e. they are continuous)
- The size of K measures the limit on the amplifying effect of the function

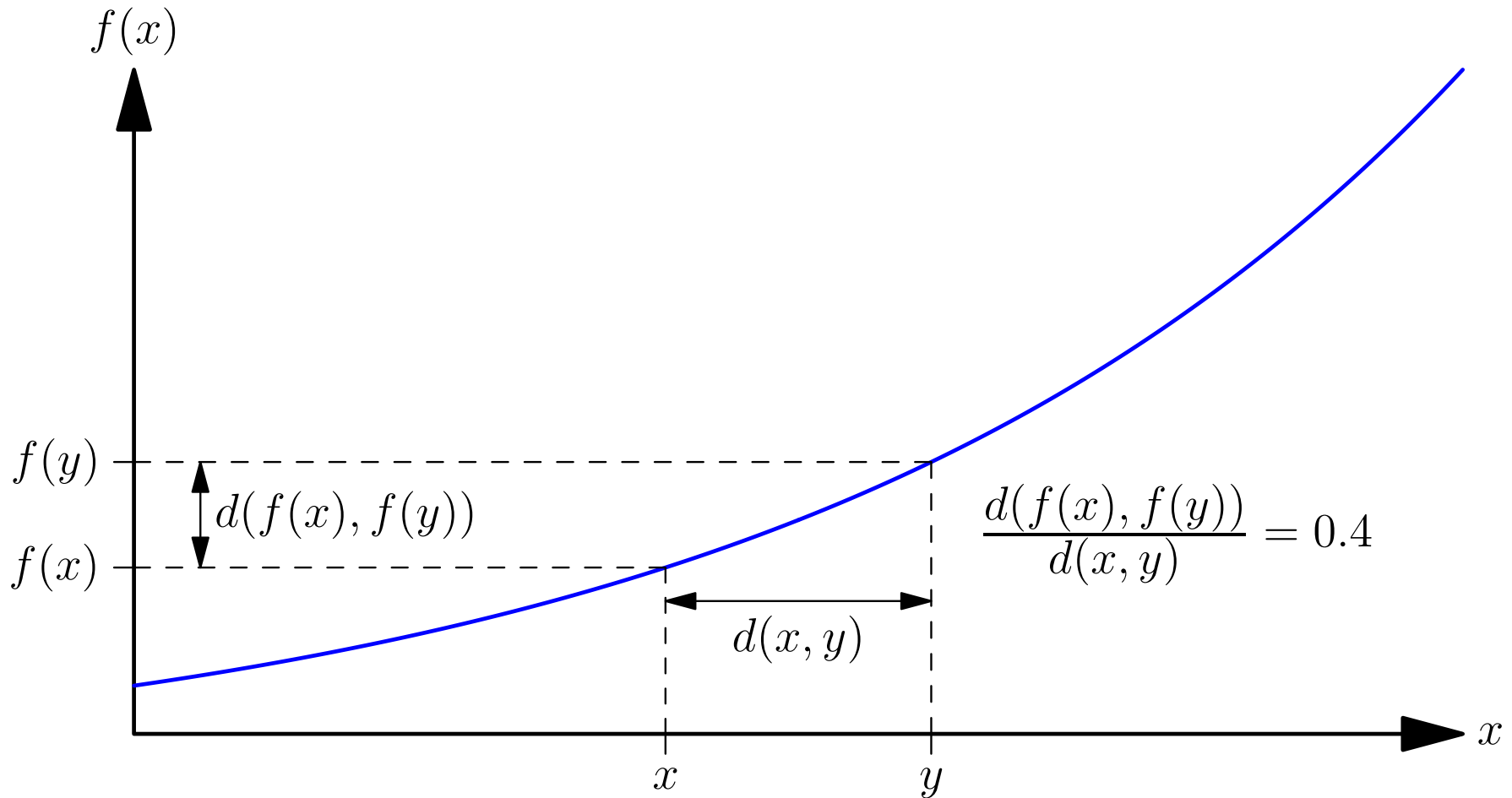
Lipschitz Continuous Functions



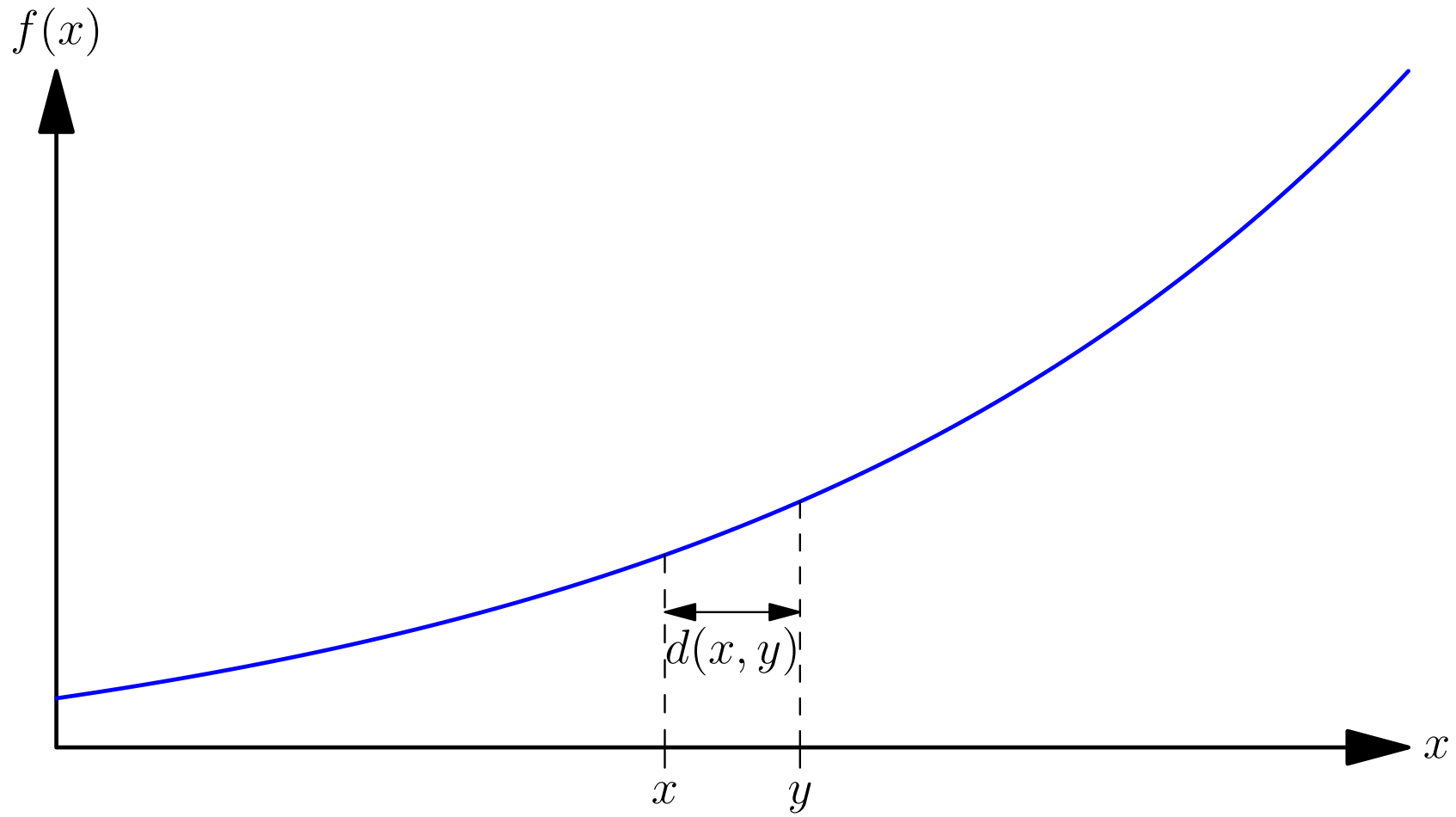
Lipschitz Continuous Functions



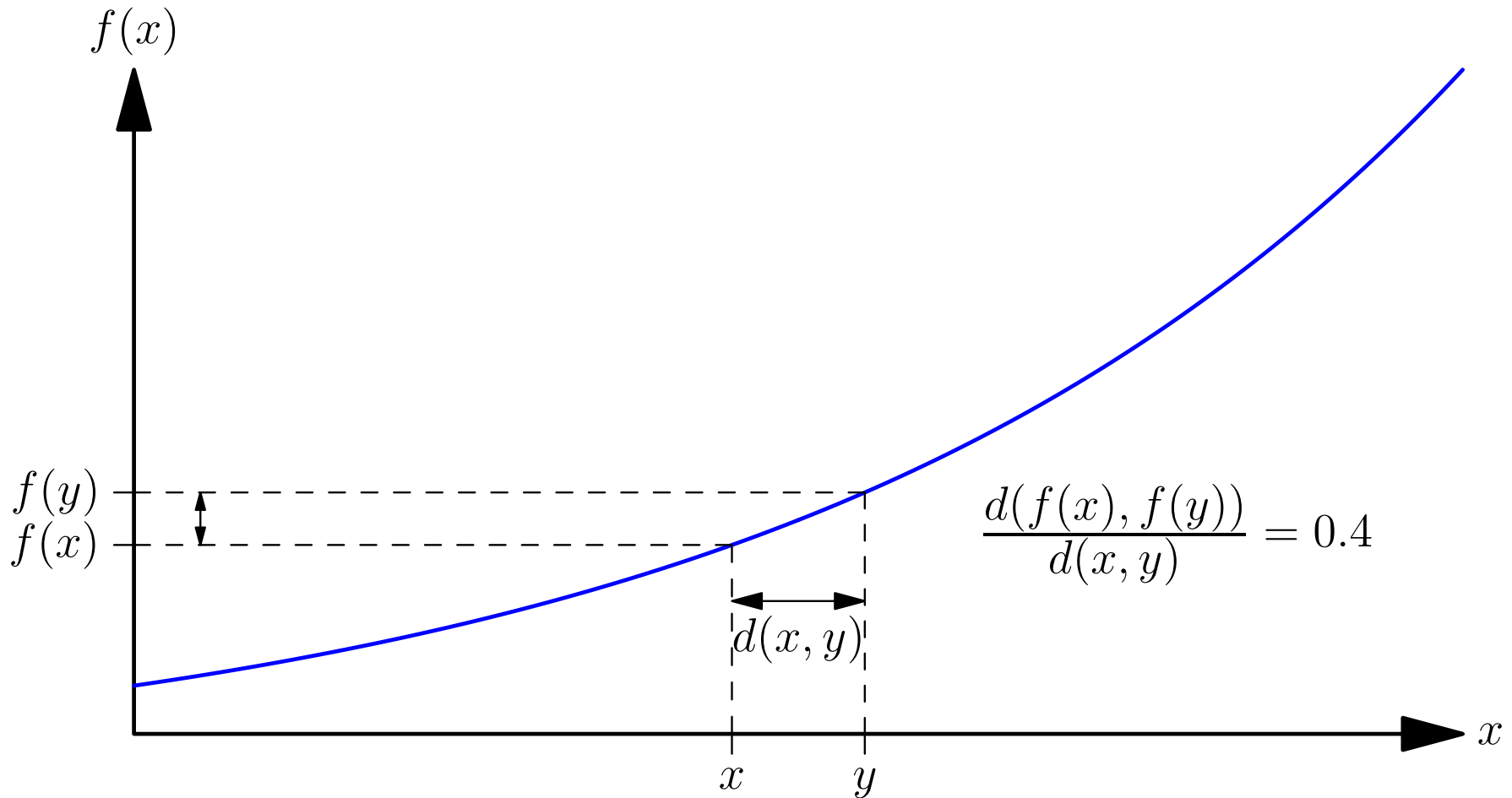
Lipschitz Continuous Functions



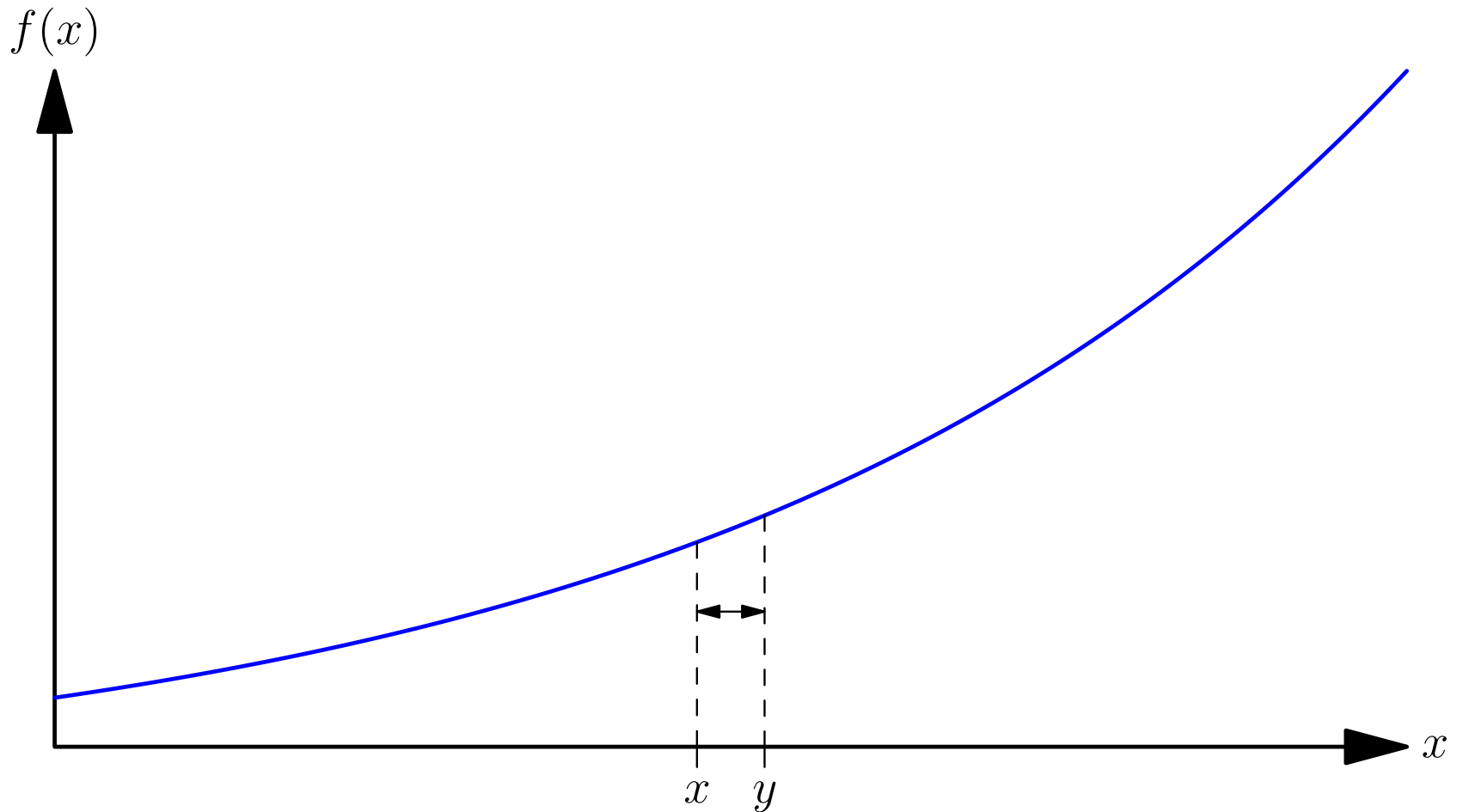
Lipschitz Continuous Functions



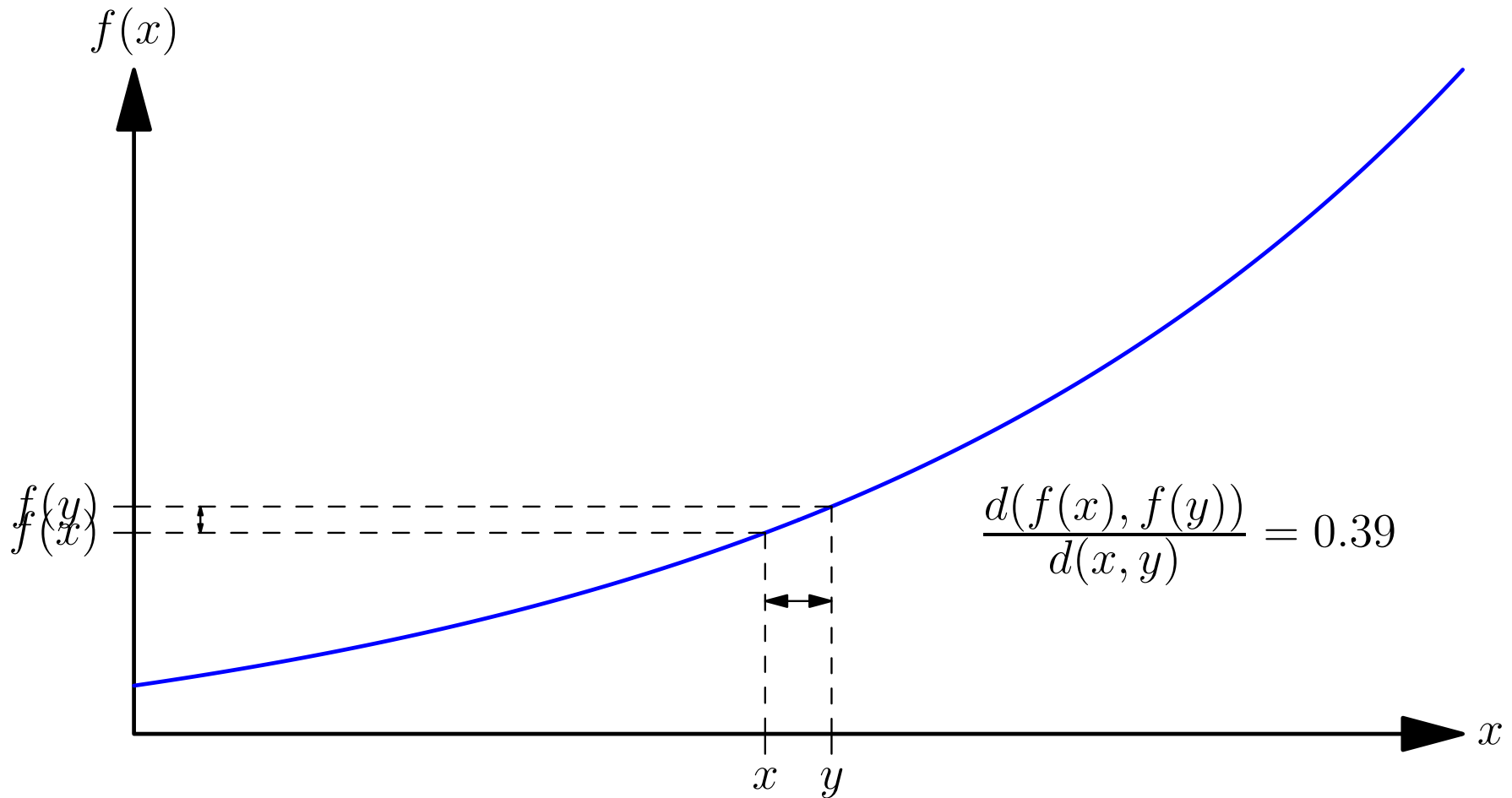
Lipschitz Continuous Functions



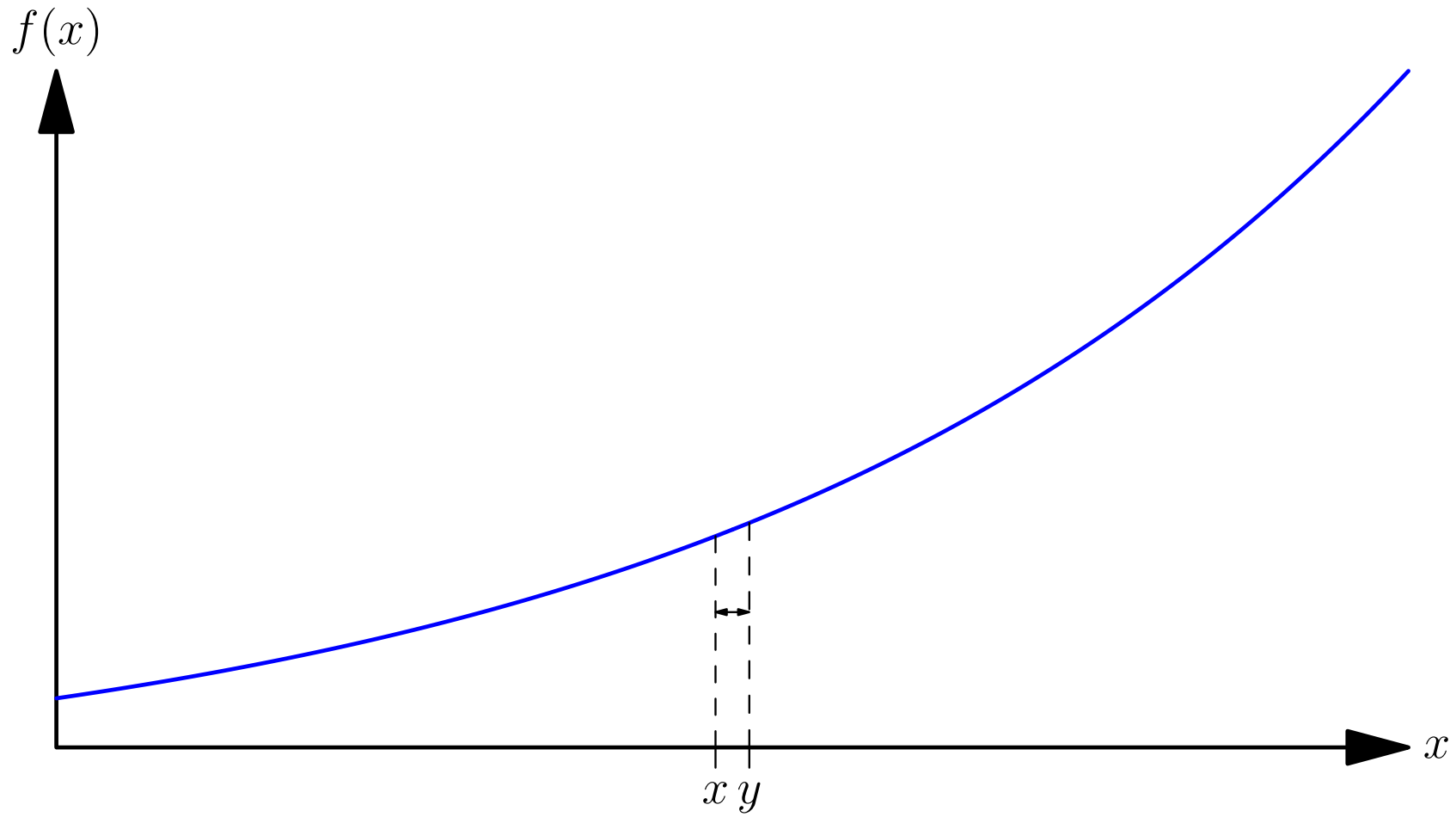
Lipschitz Continuous Functions



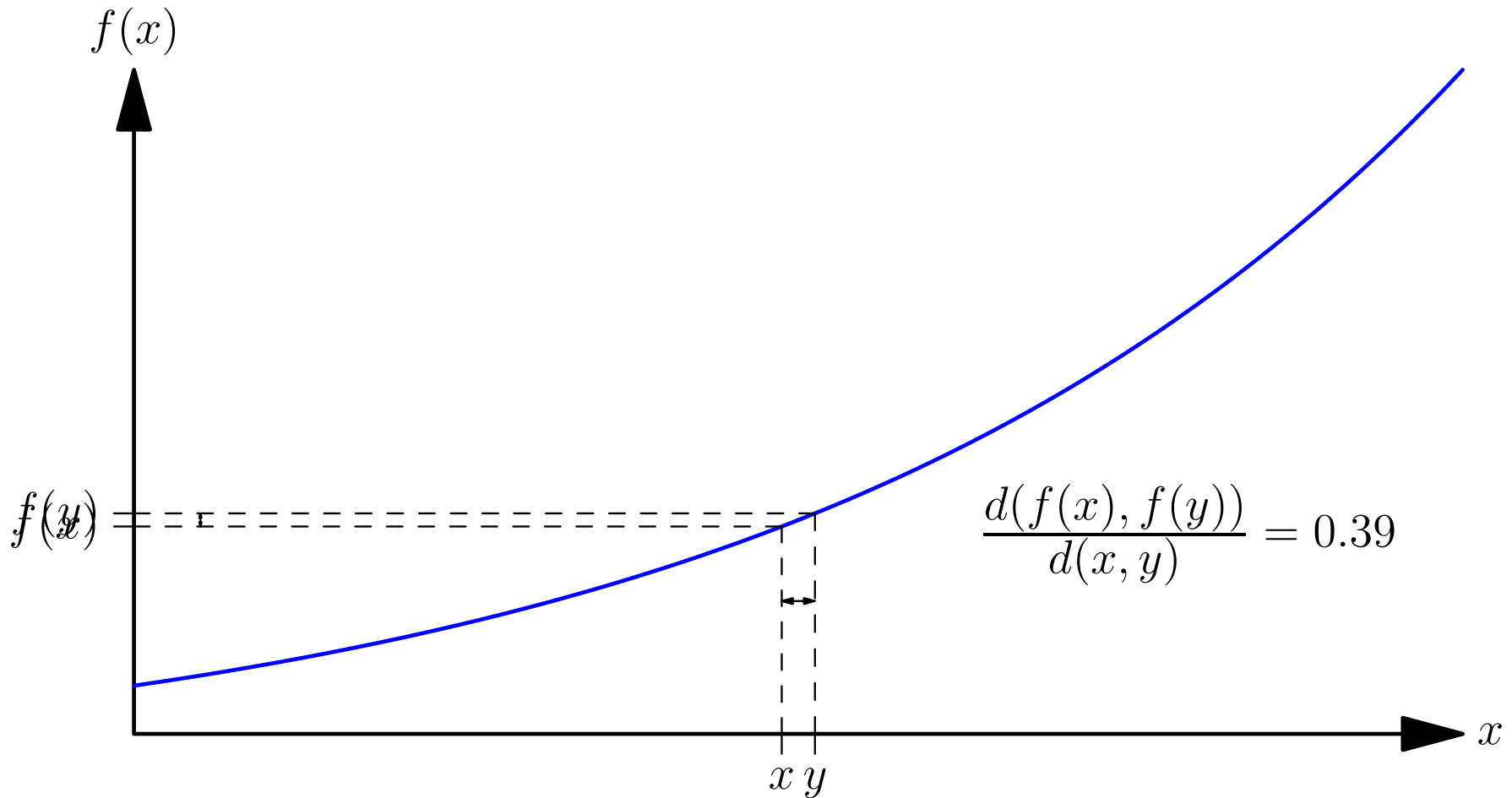
Lipschitz Continuous Functions



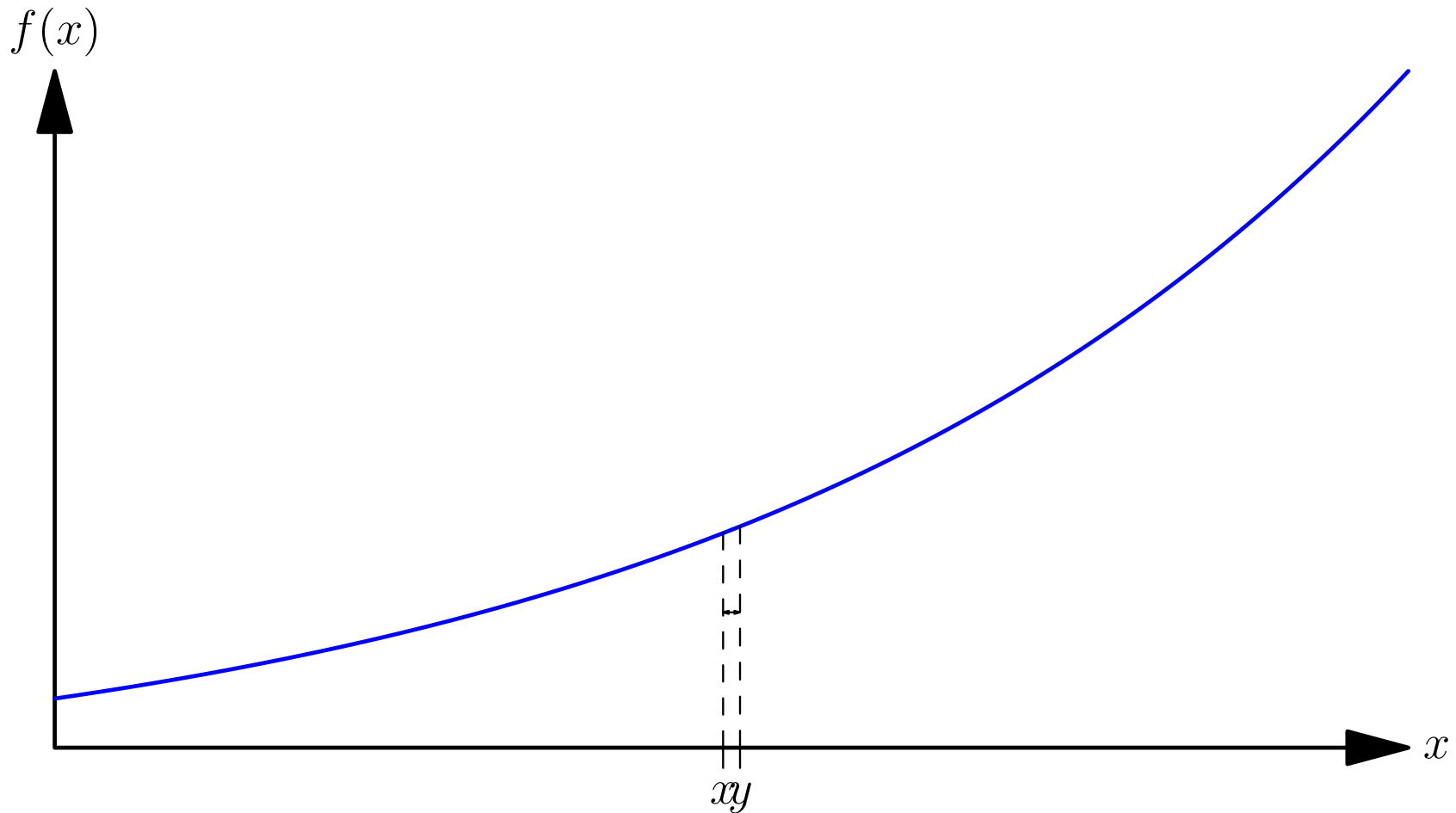
Lipschitz Continuous Functions



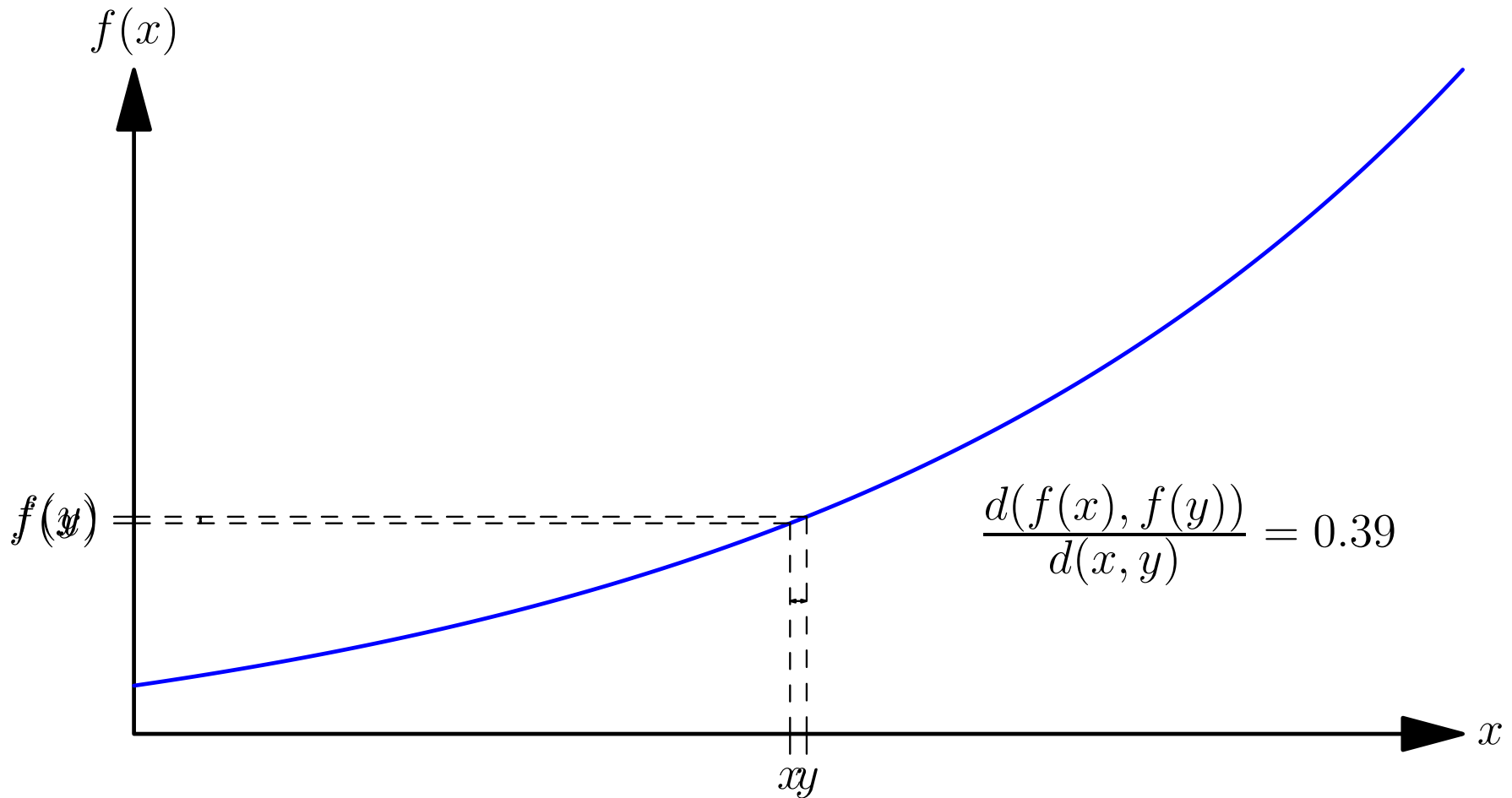
Lipschitz Continuous Functions



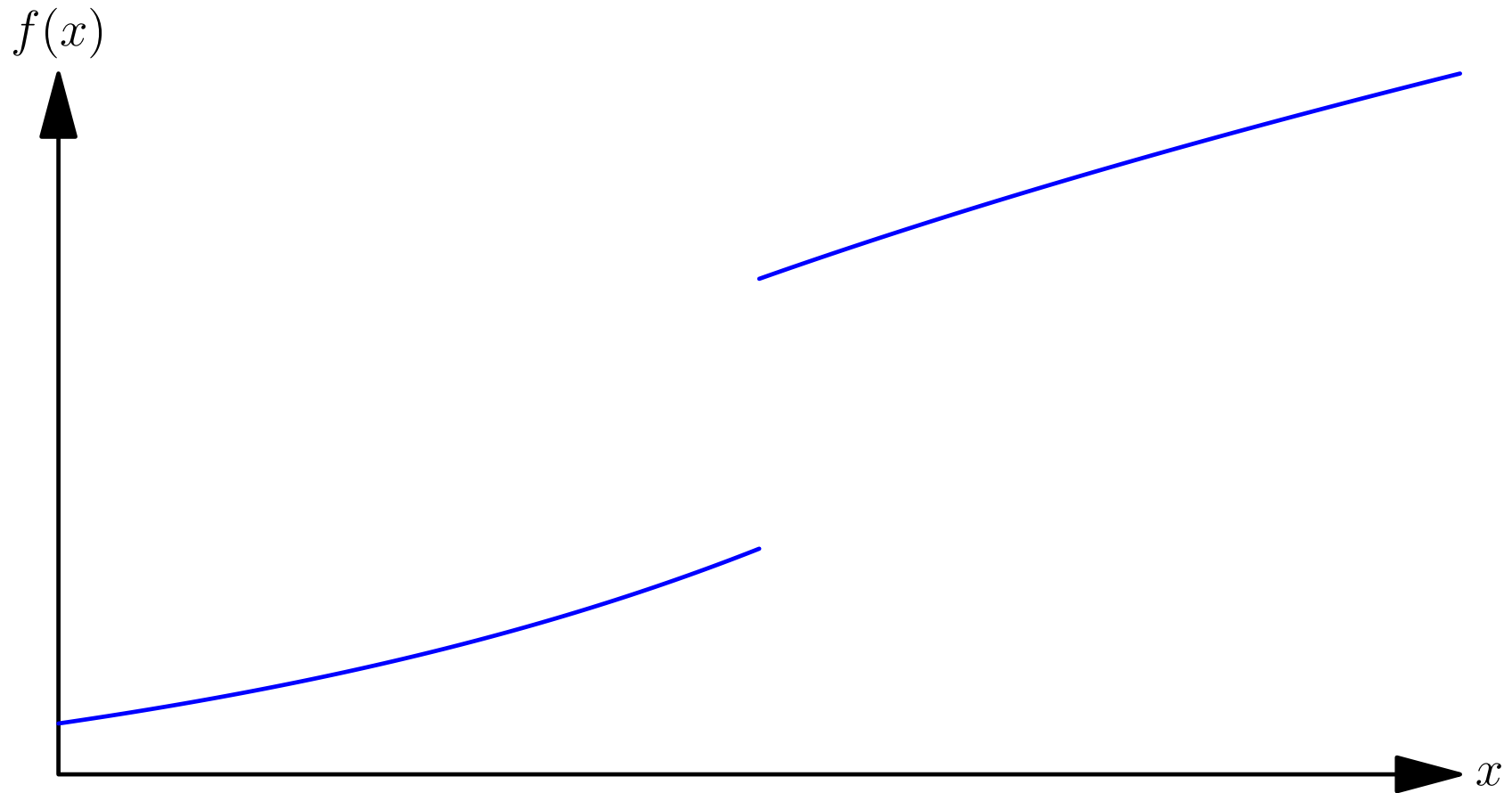
Lipschitz Continuous Functions



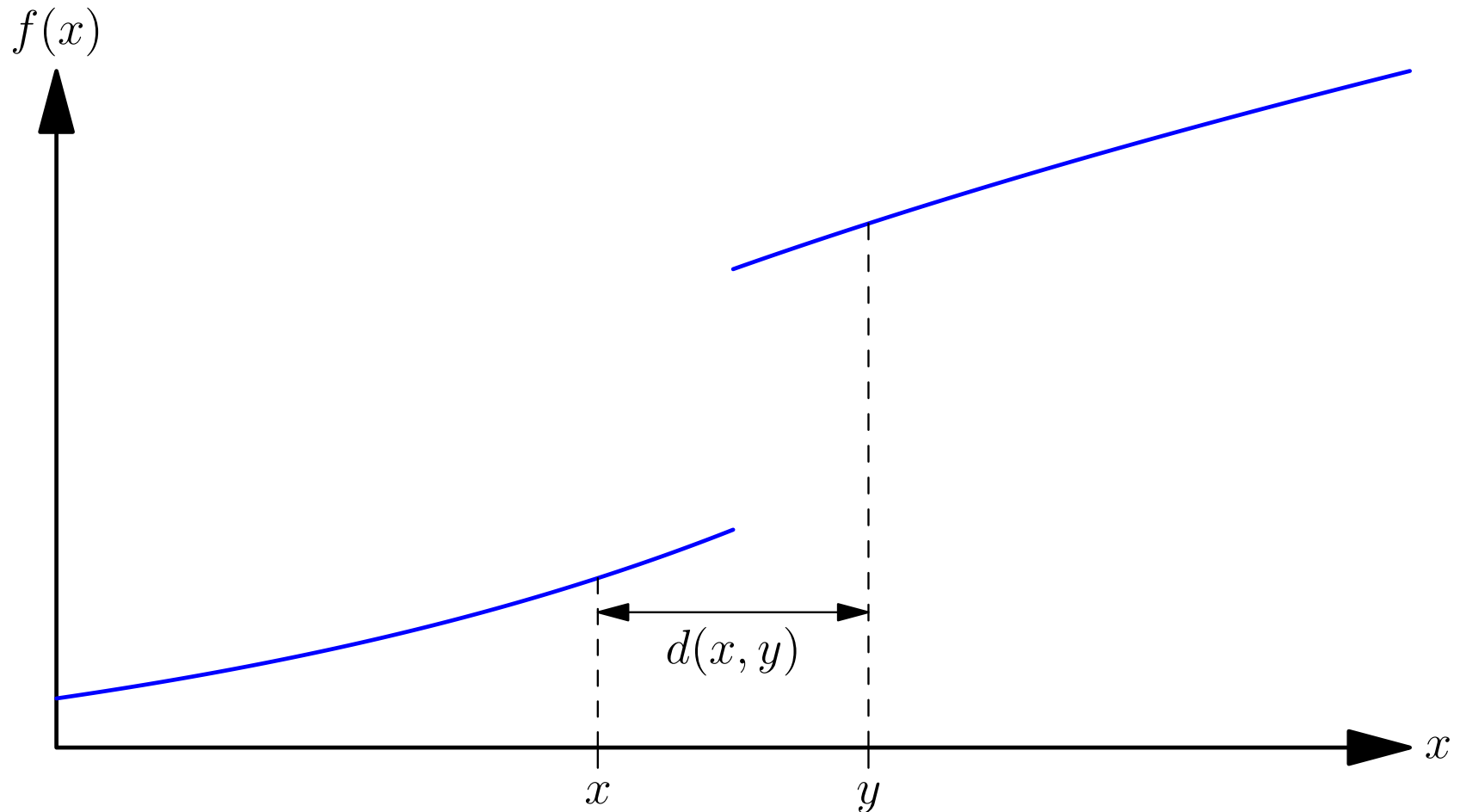
Lipschitz Continuous Functions



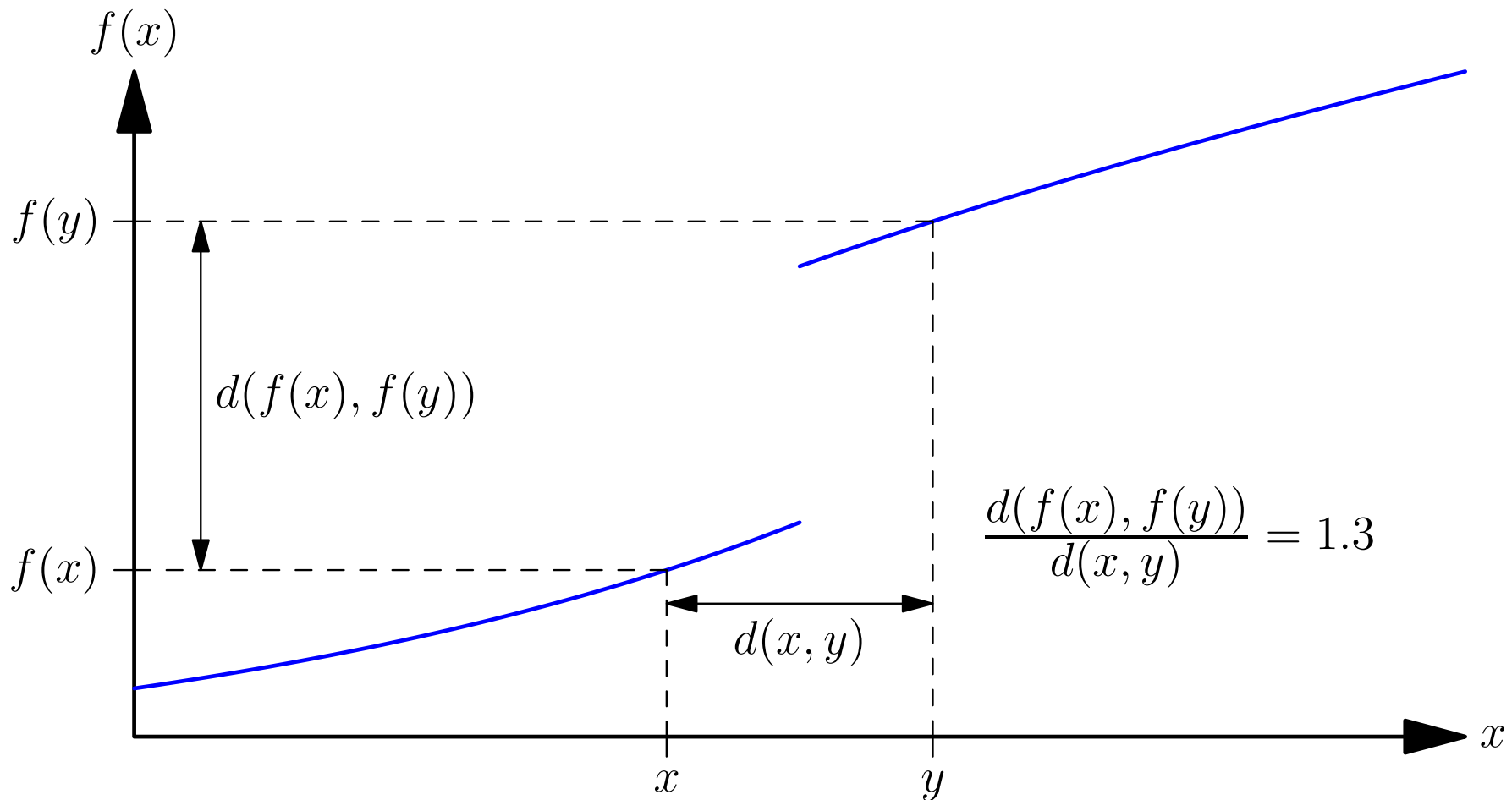
Lipschitz Continuous Functions



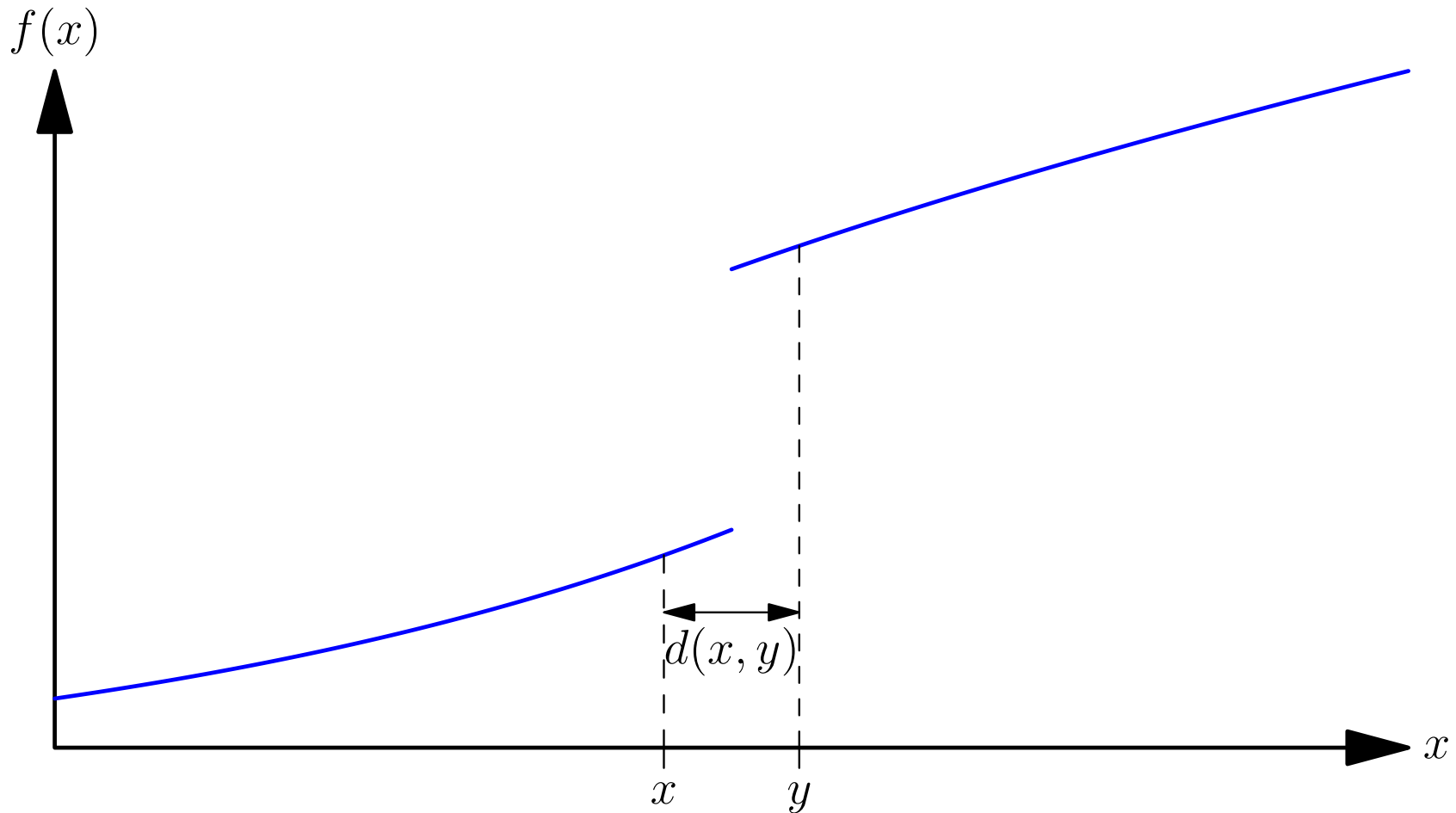
Lipschitz Continuous Functions



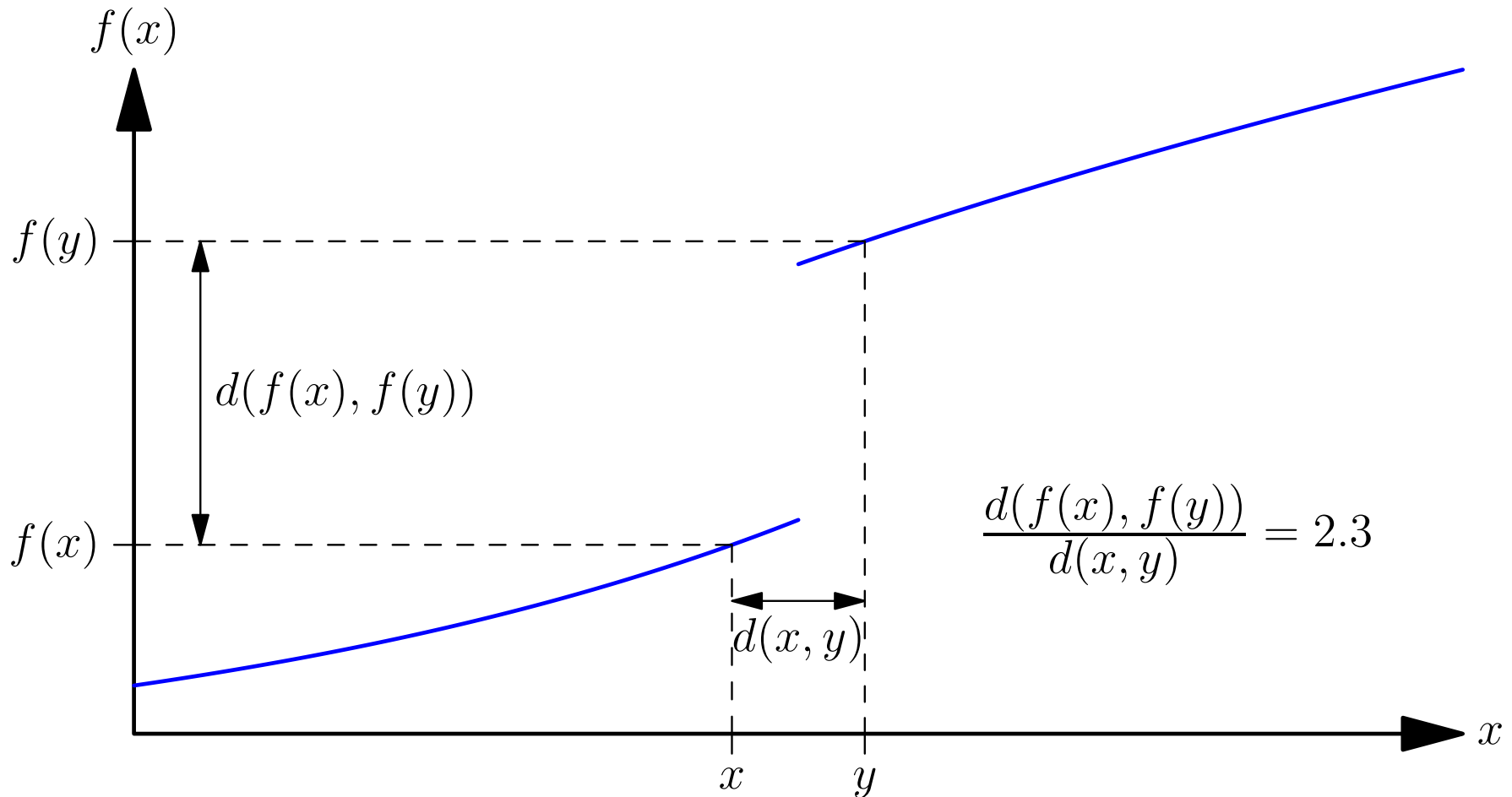
Lipschitz Continuous Functions



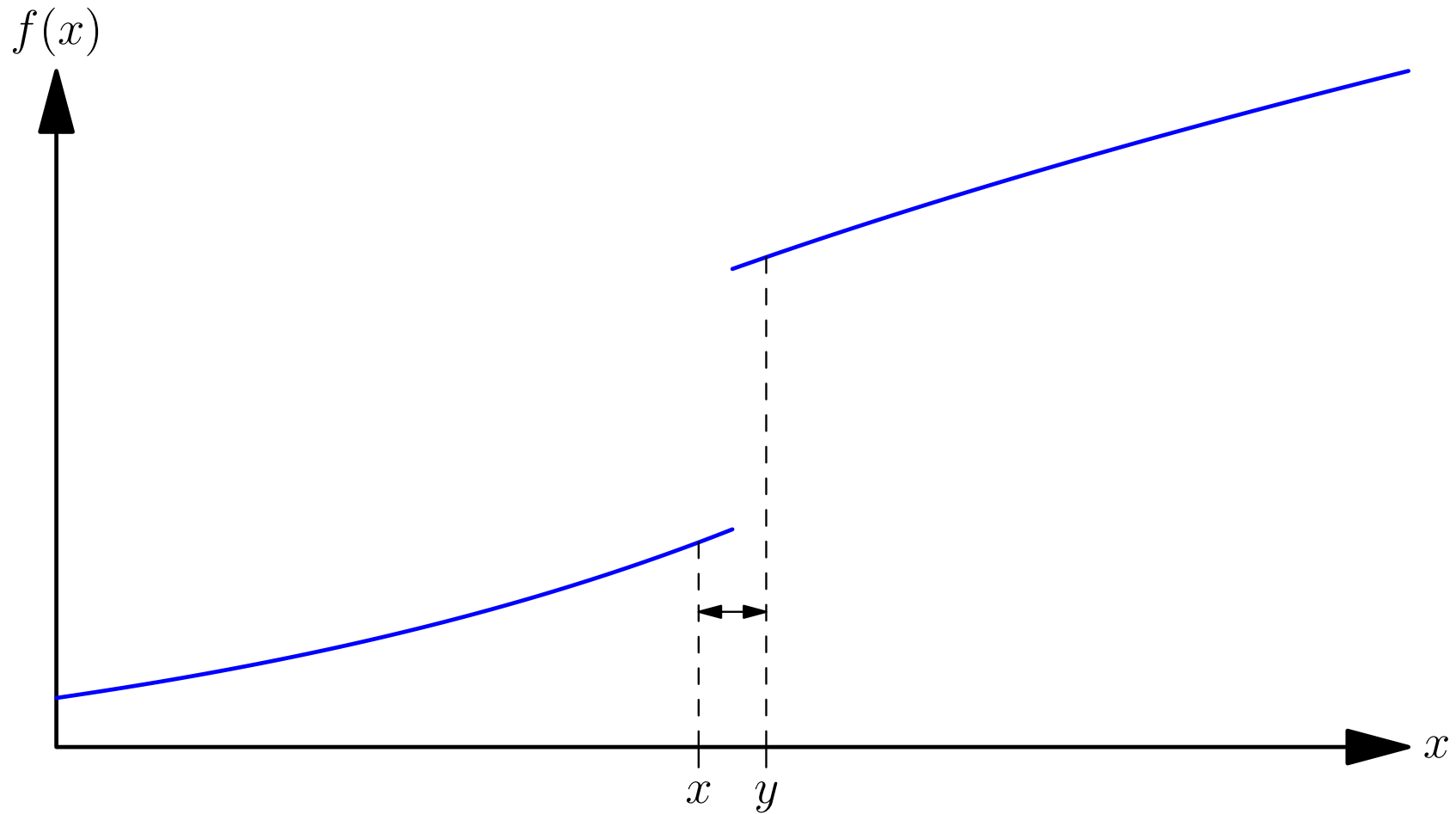
Lipschitz Continuous Functions



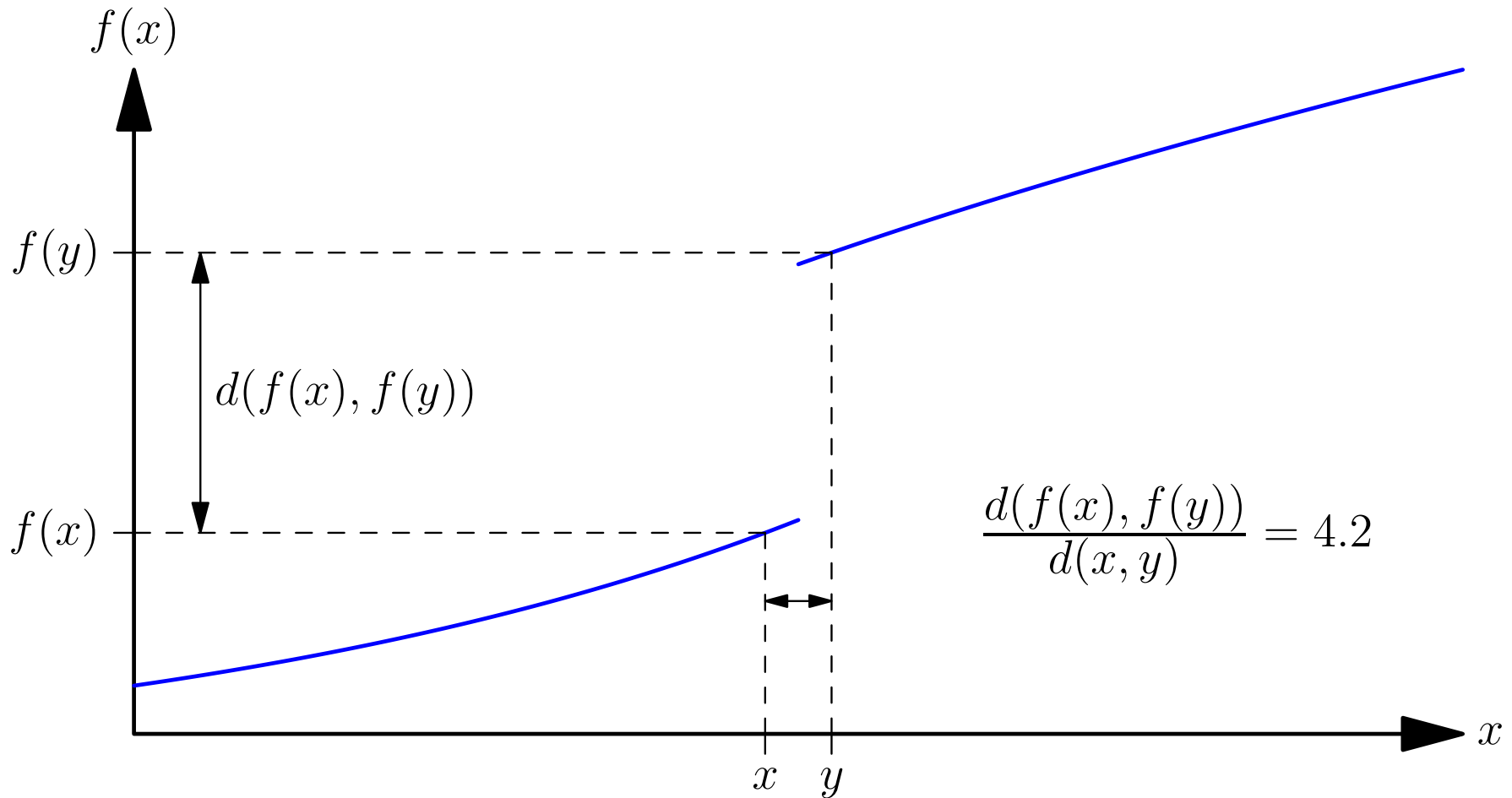
Lipschitz Continuous Functions



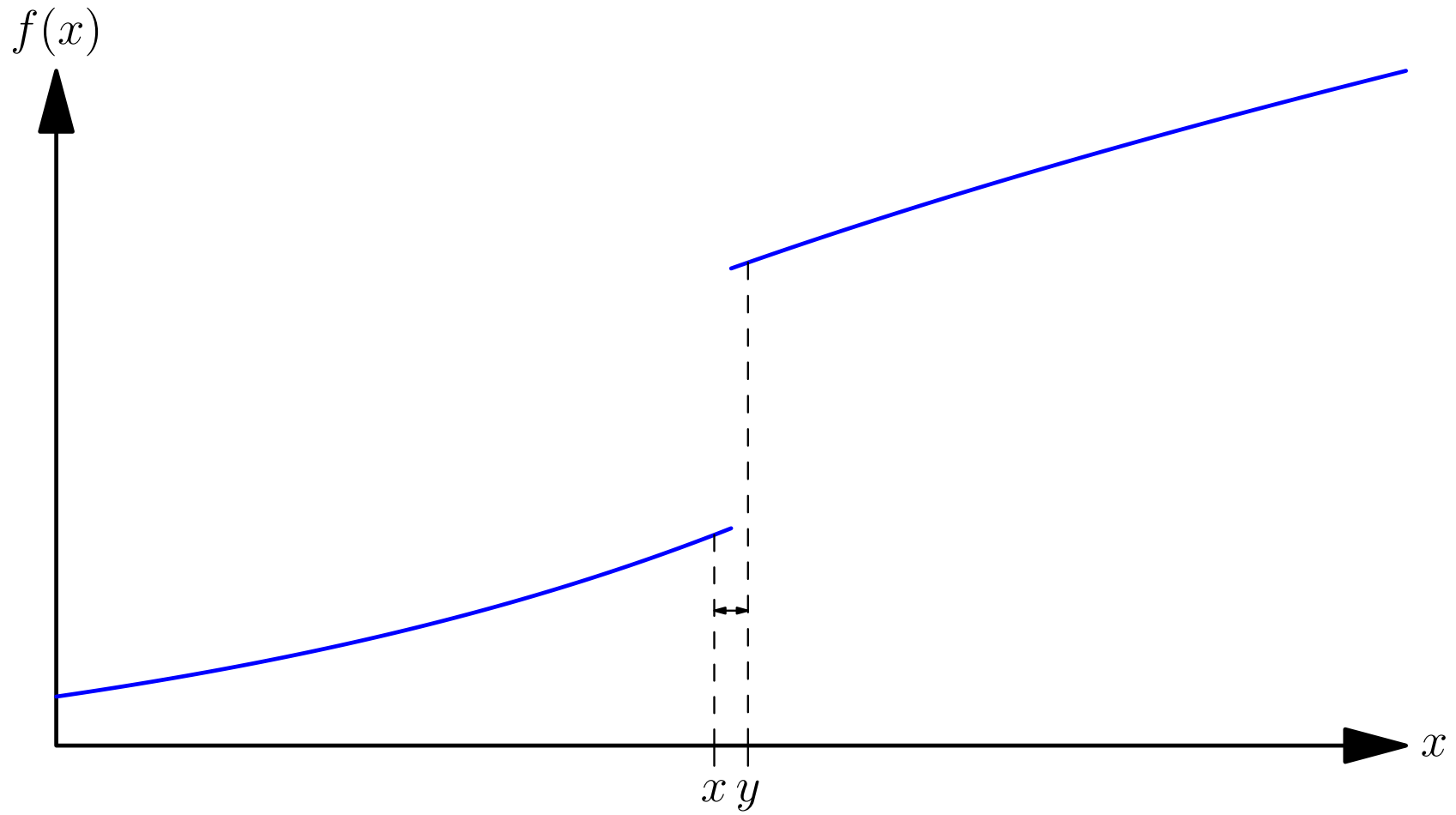
Lipschitz Continuous Functions



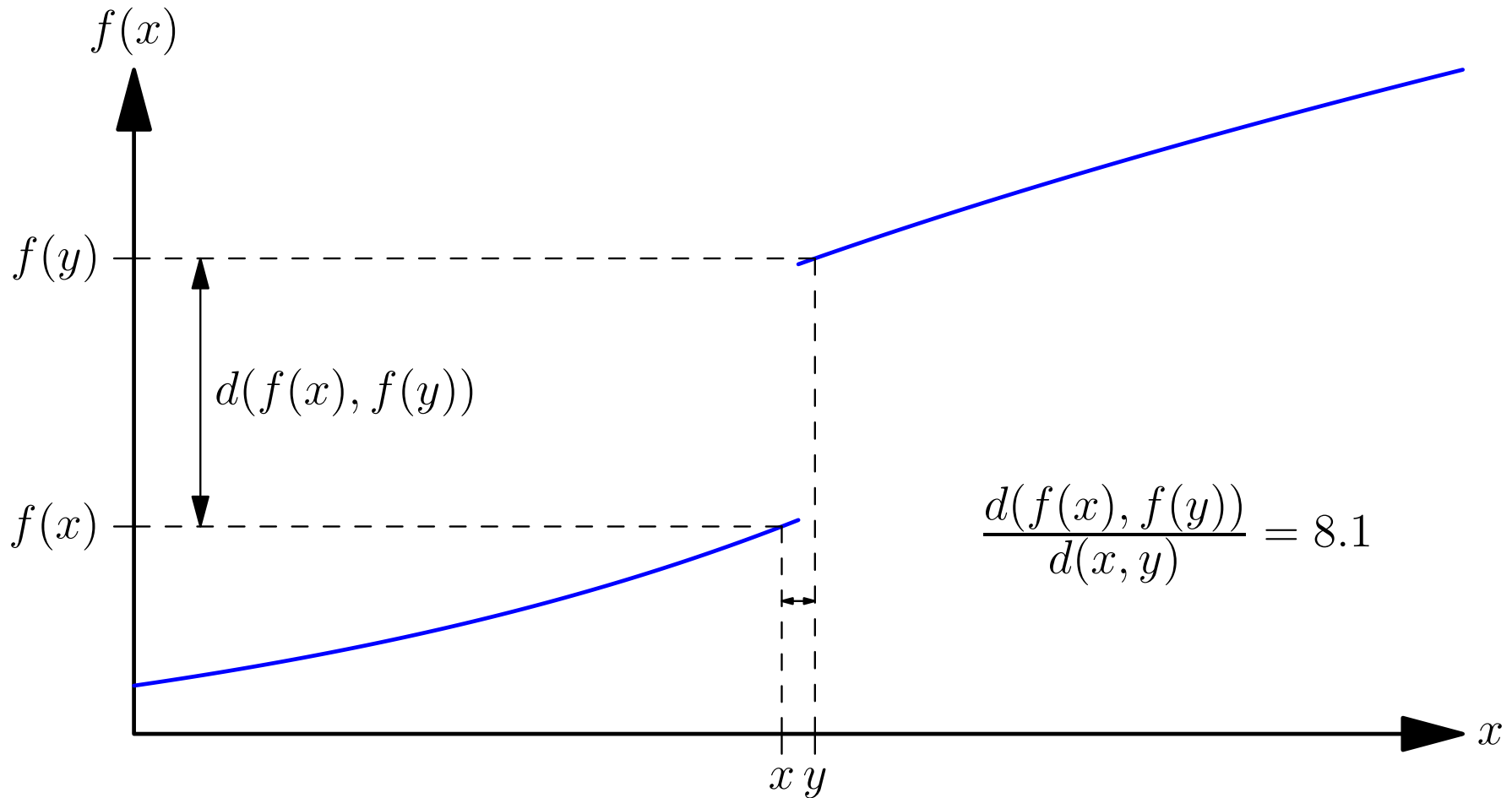
Lipschitz Continuous Functions



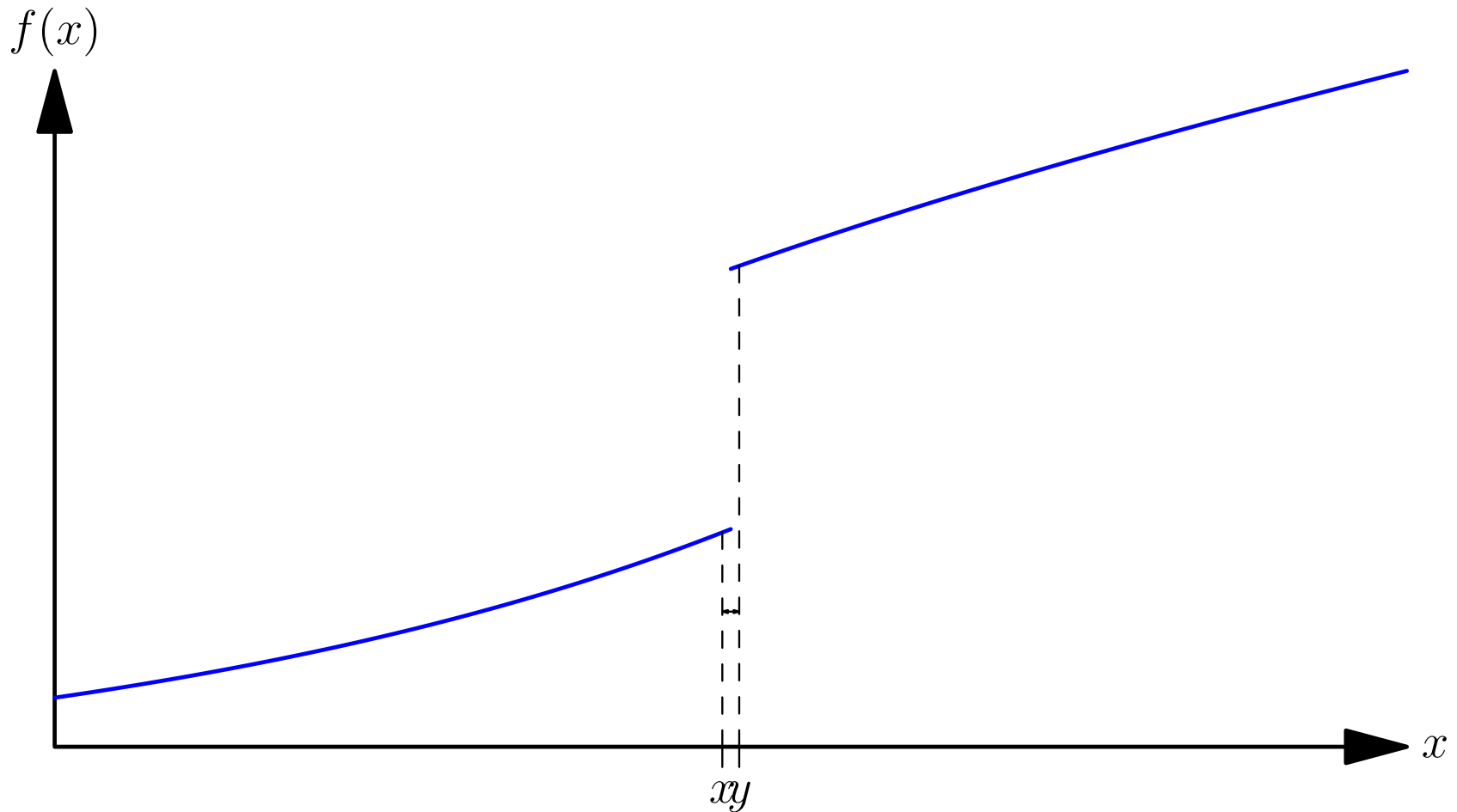
Lipschitz Continuous Functions



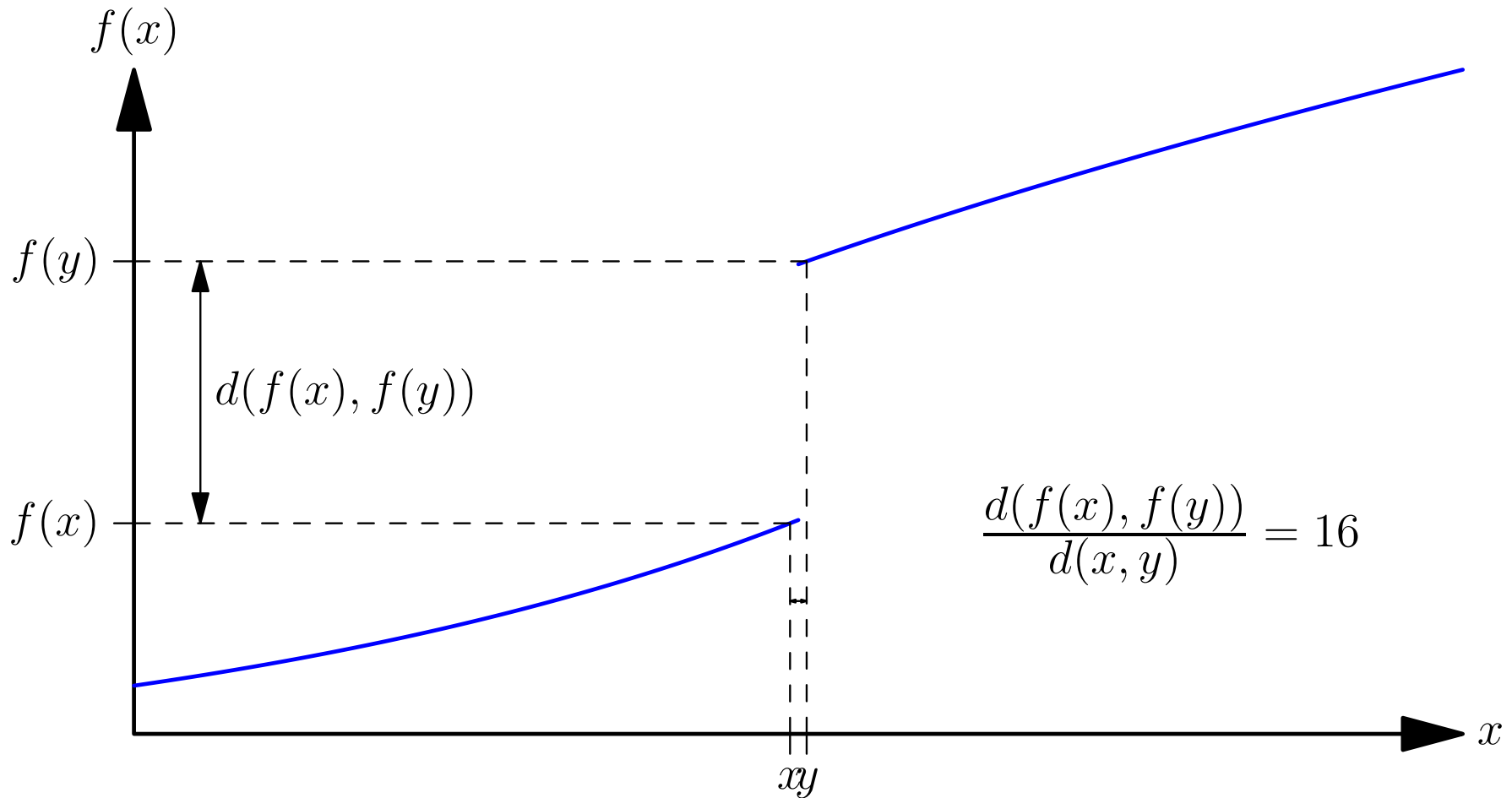
Lipschitz Continuous Functions



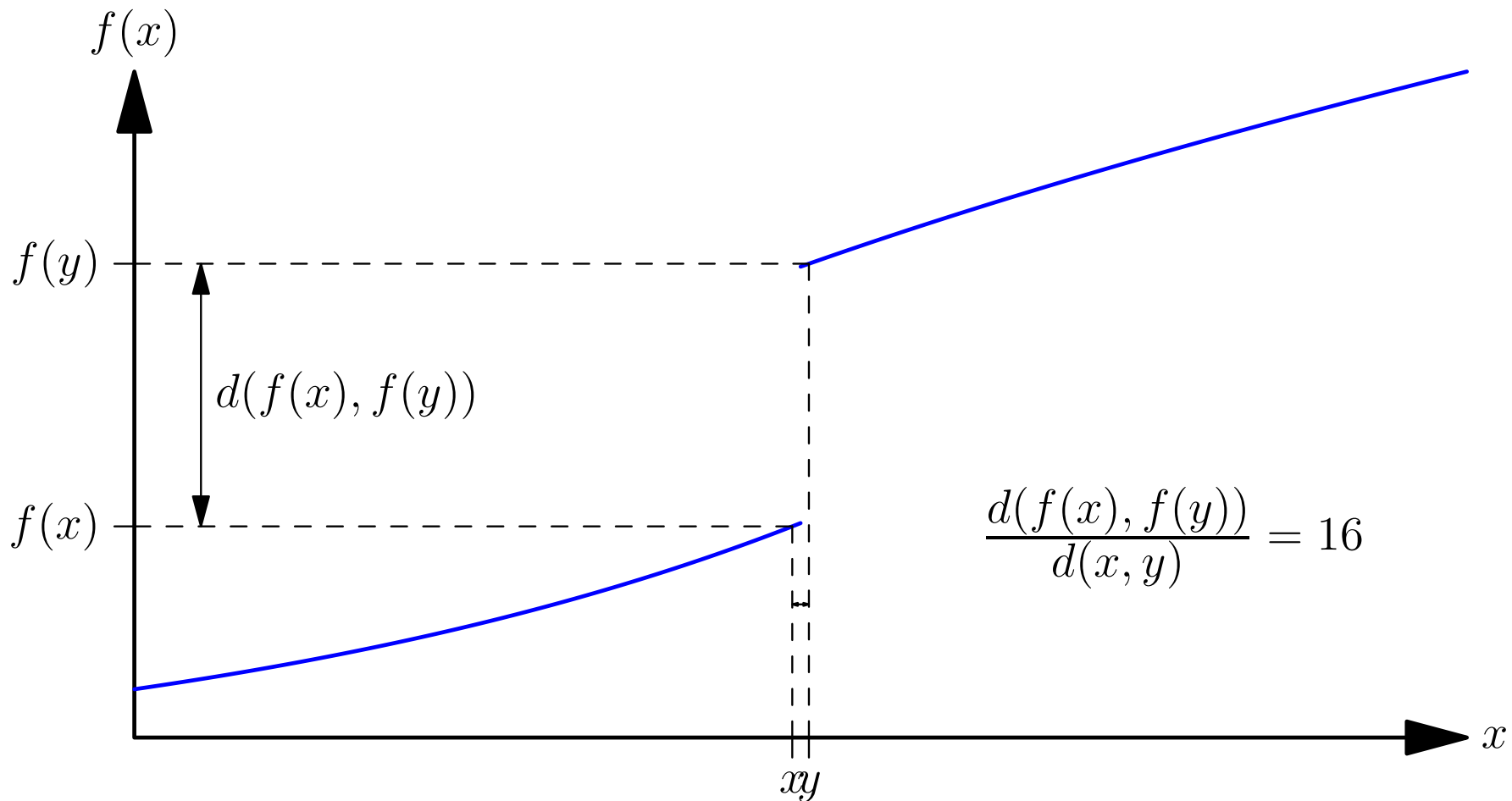
Lipschitz Continuous Functions



Lipschitz Continuous Functions



Lipschitz Continuous Functions



- Definition is applicable to any function defined in a metric space

Contractive Mappings

- An interesting class of function are those for which $K < 1$
- These are said to be contractive mappings
- A famous theorem that applies to contractive mappings is the Banach fixed-point theorem which says there exists a unique fixed point such that $f(x) = x$
- This is used for example in showing that various algorithms will converge

Contractive Mappings

- An interesting class of function are those for which $K < 1$
- These are said to be contractive mappings
- A famous theorem that applies to contractive mappings is the Banach fixed-point theorem which says there exists a unique fixed point such that $f(x) = x$
- This is used for example in showing that various algorithms will converge

Contractive Mappings

- An interesting class of function are those for which $K < 1$
- These are said to be contractive mappings
- A famous theorem that applies to contractive mappings is the Banach fixed-point theorem which says there exists a unique fixed point such that $f(x) = x$
- This is used for example in showing that various algorithms will converge

Contractive Mappings

- An interesting class of function are those for which $K < 1$
- These are said to be contractive mappings
- A famous theorem that applies to contractive mappings is the Banach fixed-point theorem which says there exists a unique fixed point such that $f(x) = x$
- This is used for example in showing that various algorithms will converge

Outline

1. Vector Spaces
2. Metrics (distances)
3. **Norms**

$$Mx=b$$

$$Mv_i = \lambda_i v_i$$

$$b=M$$

Norms

- Vector spaces are even more interesting with a notion of length
- **Norms** provide some measure of the size of a vector
- To formalise this we define the **norm** of an object $\mathbf{v}$ as $\|\mathbf{v}\|$ satisfying
 1. $\|\mathbf{v}\| > 0$ if $\mathbf{v} \neq \mathbf{0}$ (non-negativity)
 2. $\|a\mathbf{v}\| = a\|\mathbf{v}\|$ (linearity)
 3. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (triangular inequality)
- When some criteria aren't satisfied we have a **pseudo-norms**
- Norms provide a metric $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ (they are metric spaces)

Norms

- Vector spaces are even more interesting with a notion of length
- **Norms** provide some measure of the size of a vector
- To formalise this we define the **norm** of an object v as $\|v\|$ satisfying
 1. $\|v\| > 0$ if $v \neq \mathbf{0}$ (non-negativity)
 2. $\|av\| = a\|v\|$ (linearity)
 3. $\|u + v\| \leq \|u\| + \|v\|$ (triangular inequality)
- When some criteria aren't satisfied we have a **pseudo-norms**
- Norms provide a metric $d(x, y) = \|x - y\|$ (they are metric spaces)

Norms

- Vector spaces are even more interesting with a notion of length
- **Norms** provide some measure of the size of a vector
- To formalise this we define the **norm** of an object $\mathbf{v}$ as $\|\mathbf{v}\|$ satisfying
 1. $\|\mathbf{v}\| > 0$ if $\mathbf{v} \neq \mathbf{0}$ (non-negativity)
 2. $\|a\mathbf{v}\| = a\|\mathbf{v}\|$ (linearity)
 3. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (triangular inequality)
- When some criteria aren't satisfied we have a **pseudo-norms**
- Norms provide a metric $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ (they are metric spaces)

Norms

- Vector spaces are even more interesting with a notion of length
- **Norms** provide some measure of the size of a vector
- To formalise this we define the **norm** of an object v as $\|v\|$ satisfying
 1. $\|v\| > 0$ if $v \neq 0$ (non-negativity)
 2. $\|av\| = a\|v\|$ (linearity)
 3. $\|u + v\| \leq \|u\| + \|v\|$ (triangular inequality)
- When some criteria aren't satisfied we have a **pseudo-norms**
- Norms provide a metric $d(x, y) = \|x - y\|$ (they are metric spaces)

Norms

- Vector spaces are even more interesting with a notion of length
- **Norms** provide some measure of the size of a vector
- To formalise this we define the **norm** of an object v as $\|v\|$ satisfying
 1. $\|v\| > 0$ if $v \neq \mathbf{0}$ (non-negativity)
 2. $\|av\| = a\|v\|$ (linearity)
 3. $\|u + v\| \leq \|u\| + \|v\|$ (triangular inequality)
- When some criteria aren't satisfied we have a **pseudo-norms**
- Norms provide a metric $d(x, y) = \|x - y\|$ (they are metric spaces)

Norms

- Vector spaces are even more interesting with a notion of length
- **Norms** provide some measure of the size of a vector
- To formalise this we define the **norm** of an object $\mathbf{v}$ as $\|\mathbf{v}\|$ satisfying
 1. $\|\mathbf{v}\| > 0$ if $\mathbf{v} \neq \mathbf{0}$ (non-negativity)
 2. $\|a\mathbf{v}\| = a\|\mathbf{v}\|$ (linearity)
 3. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (triangular inequality)
- When some criteria aren't satisfied we have a **pseudo-norms**
- Norms provide a metric $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ (they are metric spaces)

Norms

- Vector spaces are even more interesting with a notion of length
- **Norms** provide some measure of the size of a vector
- To formalise this we define the **norm** of an object v as $\|v\|$ satisfying
 1. $\|v\| > 0$ if $v \neq \mathbf{0}$ (non-negativity)
 2. $\|av\| = a\|v\|$ (linearity)
 3. $\|u + v\| \leq \|u\| + \|v\|$ (triangular inequality)
- When some criteria aren't satisfied we have a **pseudo-norms**
- Norms provide a metric $d(x, y) = \|x - y\|$ (they are metric spaces)

Vector Norms

- The familiar vector norm is the (Euclidean) two norm

$$\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

- Other norms exist, such as the p -norm ($p \geq 1$)

$$\|\mathbf{v}\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p}$$

- Special cases include the 1-norm and the infinite norm

$$\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i| \qquad \|\mathbf{v}\|_\infty = \max_i |v_i|$$

- The 0-norm is a pseudo-norm as it does not satisfy condition 2

$$\|\mathbf{v}\|_0 = \text{number of non-zero components}$$

Vector Norms

- The familiar vector norm is the (Euclidean) two norm

$$\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

- Other norms exist, such as the p -norm ($p \geq 1$)

$$\|\mathbf{v}\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p}$$

- Special cases include the 1-norm and the infinite norm

$$\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i| \qquad \|\mathbf{v}\|_\infty = \max_i |v_i|$$

- The 0-norm is a pseudo-norm as it does not satisfy condition 2

$$\|\mathbf{v}\|_0 = \text{number of non-zero components}$$

Vector Norms

- The familiar vector norm is the (Euclidean) two norm

$$\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

- Other norms exist, such as the p -norm ($p \geq 1$)

$$\|\mathbf{v}\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p}$$

- Special cases include the 1-norm and the infinite norm

$$\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i| \qquad \|\mathbf{v}\|_\infty = \max_i |v_i|$$

- The 0-norm is a pseudo-norm as it does not satisfy condition 2

$$\|\mathbf{v}\|_0 = \text{number of non-zero components}$$

Vector Norms

- The familiar vector norm is the (Euclidean) two norm

$$\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

- Other norms exist, such as the p -norm ($p \geq 1$)

$$\|\mathbf{v}\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p}$$

- Special cases include the 1-norm and the infinite norm

$$\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i| \qquad \|\mathbf{v}\|_\infty = \max_i |v_i|$$

- The 0-norm is a pseudo-norm as it does not satisfy condition 2

$$\|\mathbf{v}\|_0 = \text{number of non-zero components}$$

Vector Norms

- The familiar vector norm is the (Euclidean) two norm

$$\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

- Other norms exist, such as the p -norm ($p \geq 1$)

$$\|\mathbf{v}\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p}$$

- Special cases include the 1-norm and the infinite norm

$$\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i| \qquad \|\mathbf{v}\|_\infty = \max_i |v_i|$$

- The 0-norm is a pseudo-norm as it does not satisfy condition 2

$$\|\mathbf{v}\|_0 = \text{number of non-zero components}$$

Matrix Norms

- We can define norms for other objects
- The norm of a matrix encodes how large the mapping is
- The Frobenius norm is defined by

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2}$$

- Many other norms exist including 1-norm, max-norm, etc.
- For square matrices, some, but not all, norms satisfy the inequality

$$\|\mathbf{AB}\| \leq \|\mathbf{A}\| \times \|\mathbf{B}\|$$

Matrix Norms

- We can define norms for other objects
- The norm of a matrix encodes how large the mapping is
- The Frobenius norm is defined by

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2}$$

- Many other norms exist including 1-norm, max-norm, etc.
- For square matrices, some, but not all, norms satisfy the inequality

$$\|\mathbf{AB}\| \leq \|\mathbf{A}\| \times \|\mathbf{B}\|$$

Matrix Norms

- We can define norms for other objects
- The norm of a matrix encodes how large the mapping is
- The Frobenius norm is defined by

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2}$$

- Many other norms exist including 1-norm, max-norm, etc.
- For square matrices, some, but not all, norms satisfy the inequality

$$\|\mathbf{AB}\| \leq \|\mathbf{A}\| \times \|\mathbf{B}\|$$

Matrix Norms

- We can define norms for other objects
- The norm of a matrix encodes how large the mapping is
- The Frobenius norm is defined by

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2}$$

- Many other norms exist including 1-norm, max-norm, etc.
- For square matrices, some, but not all, norms satisfy the inequality

$$\|\mathbf{AB}\| \leq \|\mathbf{A}\| \times \|\mathbf{B}\|$$

Matrix Norms

- We can define norms for other objects
- The norm of a matrix encodes how large the mapping is
- The Frobenius norm is defined by

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2}$$

- Many other norms exist including 1-norm, max-norm, etc.
- For square matrices, some, but not all, norms satisfy the inequality

$$\|\mathbf{AB}\| \leq \|\mathbf{A}\| \times \|\mathbf{B}\|$$

Compatible Norms

- A vector and matrix norm are said to be compatible if

$$\|\mathbf{M}\mathbf{v}\|_b \leq \|\mathbf{M}\|_a \times \|\mathbf{v}\|_b$$

(Spectral and Euclidean norms are compatible)

- Norms provide quick ways to bound the maximum growth of a vector under a mapping induced by the matrix
- We will see that a measure of the sensitivity of a mapping is in terms of the ratio of its maximum effect to its minimum effect on a vector
- This is known as the **conditioning**, given by $\|\mathbf{M}\| \times \|\mathbf{M}^{-1}\|$

Compatible Norms

- A vector and matrix norm are said to be compatible if

$$\|\mathbf{M}\mathbf{v}\|_b \leq \|\mathbf{M}\|_a \times \|\mathbf{v}\|_b$$

(Spectral and Euclidean norms are compatible)

- Norms provide quick ways to bound the maximum growth of a vector under a mapping induced by the matrix
- We will see that a measure of the sensitivity of a mapping is in terms of the ratio of its maximum effect to its minimum effect on a vector
- This is known as the **conditioning**, given by $\|\mathbf{M}\| \times \|\mathbf{M}^{-1}\|$

Compatible Norms

- A vector and matrix norm are said to be compatible if

$$\|\mathbf{M}\mathbf{v}\|_b \leq \|\mathbf{M}\|_a \times \|\mathbf{v}\|_b$$

(Spectral and Euclidean norms are compatible)

- Norms provide quick ways to bound the maximum growth of a vector under a mapping induced by the matrix
- We will see that a measure of the sensitivity of a mapping is in terms of the ratio of its maximum effect to its minimum effect on a vector
- This is known as the **conditioning**, given by $\|\mathbf{M}\| \times \|\mathbf{M}^{-1}\|$

Compatible Norms

- A vector and matrix norm are said to be compatible if

$$\|\mathbf{M}\mathbf{v}\|_b \leq \|\mathbf{M}\|_a \times \|\mathbf{v}\|_b$$

(Spectral and Euclidean norms are compatible)

- Norms provide quick ways to bound the maximum growth of a vector under a mapping induced by the matrix
- We will see that a measure of the sensitivity of a mapping is in terms of the ratio of its maximum effect to its minimum effect on a vector
- This is known as the **conditioning**, given by $\|\mathbf{M}\| \times \|\mathbf{M}^{-1}\|$

Why Should You Care?

- Deep learning involves multiply the input (which we can think of as a vector $\mathbf{x}$) by many layers
- In CNNs we have convolutional layers and dense layers
- The effect of applying these layers can be represented by a matrix multiplication $\mathbf{x}_n = \mathbf{L}_n \mathbf{x}_{n-1}$
- We also do other things like applying ReLU's or pooling that changes the magnitude, $\mathbf{x}_n$, of our representation
- If you are developing new architectures you want $\|\mathbf{x}_n\|$ neither to blow up or vanish
- This can be controlled by carefully choosing $\|\mathbf{L}_n\|$

Why Should You Care?

- Deep learning involves multiply the input (which we can think of as a vector $\mathbf{x}$) by many layers
- In CNNs we have convolutional layers and dense layers
- The effect of applying these layers can be represented by a matrix multiplication $\mathbf{x}_n = \mathbf{L}_n \mathbf{x}_{n-1}$
- We also do other things like applying ReLU's or pooling that changes the magnitude, $\mathbf{x}_n$, of our representation
- If you are developing new architectures you want $\|\mathbf{x}_n\|$ neither to blow up or vanish
- This can be controlled by carefully choosing $\|\mathbf{L}_n\|$

Why Should You Care?

- Deep learning involves multiply the input (which we can think of as a vector $\mathbf{x}$) by many layers
- In CNNs we have convolutional layers and dense layers
- The effect of applying these layers can be represented by a matrix multiplication $\mathbf{x}_n = \mathbf{L}_n \mathbf{x}_{n-1}$
- We also do other things like applying ReLU's or pooling that changes the magnitude, $\mathbf{x}_n$, of our representation
- If you are developing new architectures you want $\|\mathbf{x}_n\|$ neither to blow up or vanish
- This can be controlled by carefully choosing $\|\mathbf{L}_n\|$

Why Should You Care?

- Deep learning involves multiply the input (which we can think of as a vector $\mathbf{x}$) by many layers
- In CNNs we have convolutional layers and dense layers
- The effect of applying these layers can be represented by a matrix multiplication $\mathbf{x}_n = \mathbf{L}_n \mathbf{x}_{n-1}$
- We also do other things like applying ReLU's or pooling that changes the magnitude, $\mathbf{x}_n$, of our representation
- If you are developing new architectures you want $\|\mathbf{x}_n\|$ neither to blow up or vanish
- This can be controlled by carefully choosing $\|\mathbf{L}_n\|$

Why Should You Care?

- Deep learning involves multiply the input (which we can think of as a vector $\mathbf{x}$) by many layers
- In CNNs we have convolutional layers and dense layers
- The effect of applying these layers can be represented by a matrix multiplication $\mathbf{x}_n = \mathbf{L}_n \mathbf{x}_{n-1}$
- We also do other things like applying ReLU's or pooling that changes the magnitude, $\mathbf{x}_n$, of our representation
- If you are developing new architectures you want $\|\mathbf{x}_n\|$ neither to blow up or vanish
- This can be controlled by carefully choosing $\|\mathbf{L}_n\|$

Why Should You Care?

- Deep learning involves multiply the input (which we can think of as a vector $\mathbf{x}$) by many layers
- In CNNs we have convolutional layers and dense layers
- The effect of applying these layers can be represented by a matrix multiplication $\mathbf{x}_n = \mathbf{L}_n \mathbf{x}_{n-1}$
- We also do other things like applying ReLU's or pooling that changes the magnitude, $\mathbf{x}_n$, of our representation
- If you are developing new architectures you want $\|\mathbf{x}_n\|$ neither to blow up or vanish
- This can be controlled by carefully choosing $\|\mathbf{L}_n\|$

Function Norms

- Functions can also have norms, for example, if $f(x)$ is defined in some interval $\mathcal{I}$

$$\|f\|_{L_2} = \sqrt{\int_{x \in \mathcal{I}} f^2(x) dx}$$

- The L_2 vector space is the set of function where $\|f\|_{L_2} < \infty$
- The L_1 -norm is given by $\|f\|_{L_1} = \int_{x \in \mathcal{I}} |f(x)| dx$
- The infinite-norm is given by $\|f\|_{\infty} = \max_{x \in \mathcal{I}} |f(x)|$

Function Norms

- Functions can also have norms, for example, if $f(x)$ is defined in some interval $\mathcal{I}$

$$\|f\|_{L_2} = \sqrt{\int_{x \in \mathcal{I}} f^2(x) dx}$$

- The L_2 vector space is the set of function where $\|f\|_{L_2} < \infty$
- The L_1 -norm is given by $\|f\|_{L_1} = \int_{x \in \mathcal{I}} |f(x)| dx$
- The infinite-norm is given by $\|f\|_{\infty} = \max_{x \in \mathcal{I}} |f(x)|$

Function Norms

- Functions can also have norms, for example, if $f(x)$ is defined in some interval $\mathcal{I}$

$$\|f\|_{L_2} = \sqrt{\int_{x \in \mathcal{I}} f^2(x) dx}$$

- The L_2 vector space is the set of function where $\|f\|_{L_2} < \infty$
- The L_1 -norm is given by $\|f\|_{L_1} = \int_{x \in \mathcal{I}} |f(x)| dx$
- The infinite-norm is given by $\|f\|_{\infty} = \max_{x \in \mathcal{I}} |f(x)|$

Function Norms

- Functions can also have norms, for example, if $f(x)$ is defined in some interval $\mathcal{I}$

$$\|f\|_{L_2} = \sqrt{\int_{x \in \mathcal{I}} f^2(x) dx}$$

- The L_2 vector space is the set of function where $\|f\|_{L_2} < \infty$
- The L_1 -norm is given by $\|f\|_{L_1} = \int_{x \in \mathcal{I}} |f(x)| dx$
- The infinite-norm is given by $\|f\|_{\infty} = \max_{x \in \mathcal{I}} |f(x)|$

Summary

- Vector spaces with a distance (metric spaces) and vector spaces with a norm (normed vector spaces) are interesting objects
- They allow you to define a topology (open/closed sets, etc.)
- You can build up ideas about connectedness, continuity, contractive maps, fixed-point theorems, . . .
- For the most part we are going to consider an even more powerful vector space that has an inner-product defined

Summary

- Vector spaces with a distance (metric spaces) and vector spaces with a norm (normed vector spaces) are interesting objects
- They allow you to define a topology (open/closed sets, etc.)
- You can build up ideas about connectedness, continuity, contractive maps, fixed-point theorems, . . .
- For the most part we are going to consider an even more powerful vector space that has an inner-product defined

Summary

- Vector spaces with a distance (metric spaces) and vector spaces with a norm (normed vector spaces) are interesting objects
- They allow you to define a topology (open/closed sets, etc.)
- You can build up ideas about connectedness, continuity, contractive maps, fixed-point theorems, . . .
- For the most part we are going to consider an even more powerful vector space that has an inner-product defined

Summary

- Vector spaces with a distance (metric spaces) and vector spaces with a norm (normed vector spaces) are interesting objects
- They allow you to define a topology (open/closed sets, etc.)
- You can build up ideas about connectedness, continuity, contractive maps, fixed-point theorems, . . .
- For the most part we are going to consider an even more powerful vector space that has an inner-product defined